
The Saturation Regularity: When Concept-Aligned Supervision Stops Helping Converged EEG Classifiers

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Abstract

1 A standard way to inject prior knowledge into a neural network is an auxiliary loss
2 that pulls its features toward a known concept direction. We report that this strategy
3 fails once the network has already converged on the concept from task labels alone.
4 On FACED, a nine-class EEG emotion benchmark, twenty-five such recipes—
5 knowledge distillation, representational similarity, contrastive, and topographic
6 losses—applied to a converged classifier improve accuracy in none of the cases
7 and significantly reduce it in fifteen. The same intervention flips sign with baseline
8 strength: a valence loss that helps a weak backbone (LaBraM, +0.057 balanced
9 accuracy, $p=0.019$) significantly hurts a strong one (-0.031 , $p=0.005$). We name
10 this the *saturation regularity*: once task labels have driven the classifier onto the
11 target concept, added supervision deforms an already-converged representation
12 without improving generalization—training accuracy stays at ceiling while test
13 accuracy falls. Saturation has a constructive corollary. The class-mean structure of
14 the features is shared across seeds, whereas the within-class residual is seed-specific
15 and load-bearing. Ensembling across this residual diversity, rather than supervising
16 the saturated basin, reaches a new FACED state of the art at **0.6948** balanced
17 accuracy (+10.5% over the prior best). The saturation regularity itself replicates on
18 a second EEG dataset, SEED-V, where the same auxiliary supervision is likewise
19 within seed noise of the baseline ($\Delta=+0.0025$). To identify what the network
20 has saturated on, we extract the concept explicitly as a one-dimensional valence
21 direction (the *V-axis*) from nine short stories passed through a language model.
22 This direction predicts movie-review sentiment zero-shot (AUC 0.832), tracks the
23 cohort-averaged EEG response to affective videos at the emotion-category level
24 ($r=0.87$), and is recovered without supervision by converged EEG classifiers. The
25 transfer is valence-specific but not V-axis-specific—a generic valence axis performs
26 identically—and population-level rather than within-subject. Saturation is thus a
27 property of the converged representation, not of any particular concept or source
28 model.

29 1 Introduction

30 Auxiliary losses that pull a network’s features toward a known concept direction are a standard
31 way to inject prior knowledge. We report a regime in which they systematically backfire. Take
32 an EEG emotion classifier that has already converged, and add a loss aligning its features with a
33 valence direction: across twenty-five such recipes on FACED [Chen et al., 2023], none improve
34 balanced accuracy and fifteen significantly reduce it. The same loss that helps a weak backbone hurts
35 a strong one. We call this the *saturation regularity*: once task labels alone have driven the network
36 onto the target concept, further supervision can only deform an already-converged representation,
37 without improving generalization. The finding is doubly useful. It explains a widespread failure

mode of concept-alignment supervision, and, by locating where the recoverable gain does live—in the seed-specific residual that supervision cannot exploit—it yields a new state of the art through ensembling rather than supervising.

To make “the concept the network has saturated on” concrete, we extract it explicitly. Pass nine short emotion-evocative sentences through a language model, average the late-layer hidden states class by class, and take the first principal component; the resulting *valence axis* (V-axis) predicts movie-review sentiment at AUC 0.832 zero-shot, tracks the cohort-averaged EEG response of 123 subjects at the emotion-category level ($r=0.87$), and is rediscovered without supervision by every reasonably-strong EEG classifier we tested. Recent work shows that language models’ internal features align with human cognition across high-level concepts [Kim et al., 2018, Arditi et al., 2024]; we use the V-axis as a *probe* of what brain-decoding networks encode, not as a training signal—which, as the saturation regularity shows, a converged network cannot benefit from.

A nine-sentence axis. The nine stories are one per FACED [Chen et al., 2023] class; the language model is Qwen3-4B, chosen as the lead because it sits at the centre of the alignment manifold in our 14-LLM sweep, with the highest per-stimulus alignment to a behavioural valence reference (the recipe itself is model-agnostic, see Appendix S3). The SST-2 benchmark grounds the AUC 0.832 headline: the V-axis projection alone is within 0.005 of a 5k-example supervised logistic regression [Socher et al., 2013]. And the direction is not idiosyncratic to any one model: across the 14 language models from 560M to 32B parameters, the V-axis converges on essentially the same direction (within Qwen3, off-diagonal $r=+0.995$ over 15 size-pairs).

The same axis lives in the brain. The $r=0.87$ above unpacks as follows. A ridge regression on 160 channel-band features of the FACED cohort EEG predicts the V-axis position of each of the 28 stimuli; the regressor uses leave-one-stimulus-out cross-validation, no per-subject fitting, no auxiliary supervision. The Qwen3-4B V-axis itself reaches $r=0.80$ ($p < 10^{-5}$) on this ridge; a CLIP-text variant of the V-axis built from the same 28 stimulus descriptions tightens this to $r=0.87$ ($p < 10^{-9}$). The third claim — that EEG networks rediscover the V-axis without ever being shown it — is quantified across 36 FACED-9 checkpoints trained on the 9 emotion labels alone: per-checkpoint V-axis encoding strength in the class-mean subspace predicts balanced accuracy at $r=+0.885$ ($p=7.8\times 10^{-13}$, Figure 1). The same valence direction is recovered from inside language models and from inside human electrophysiology, by independent procedures.

Yet supervision fails. The natural way to use such an alignment is an auxiliary loss pulling the EEG network’s penultimate features toward the V-axis. Across 25 such recipes—knowledge distillation [Hinton et al., 2015], representational similarity [Kriegeskorte et al., 2008], contrastive [Khosla et al., 2020, van den Oord et al., 2018], PEFT adapters [Hu et al., 2022], and topographic losses [Padakanti et al., 2025]—none help: fifteen produce statistically significant decrements and zero produce gains, five families showing *monotonic destruction*. The effect flips sign with baseline strength (no measurable effect at $\text{BACC} \leq 0.62$, a significant negative at $\text{BACC} \geq 0.66$). We call this *saturation*: distinct from overfitting, it is what happens once task labels alone have driven the network onto the target direction, so added supervision moves the class-mean component further along the axis ($|\Delta r|$ up to $+0.36$) while test accuracy falls and training accuracy stays at ceiling.

Where the gain lives. The same diagnosis locates the recoverable gain in the within-class residual that supervision cannot reach: the class-mean basin is shared across seeds, while the residual is seed-specific and load-bearing. Ensembling across seeds averages out residual noise without disturbing the basin—per-checkpoint residual encoding predicts leave-one-out contribution at $r=+0.74$, and a 10-checkpoint ensemble reaches FACED-9 SOTA at **0.6948** (val-selected single **0.6755**), a $+10.5\%$ improvement over EMOD [Chen et al., 2025]. Replacing the language-derived KD prototypes with random orthonormal vectors changes BACC by ≤ 0.003 : the gain is the *shape* of the loss, not the *content* of the prototypes, so the V-axis is a probe of what the network saturated on, not the supervised signal.

Contributions. (1) A nine-sentence *V-axis probe*: a one-dimensional valence direction extracted from 9 language-model emotion stories that converges across 14 LLMs, predicts text sentiment zero-shot, predicts cohort EEG response, and is rediscovered without supervision by 36 EEG classifiers (§6–§8). (2) The *saturation regularity*: across 25 representation-alignment recipes on FACED-9, 15 produce significant decrements and 0 produce gains, with monotonic destruction in 5 families and a sharp transition in the $[0.62, 0.66]$ BACC band; the loss moves the class-mean component

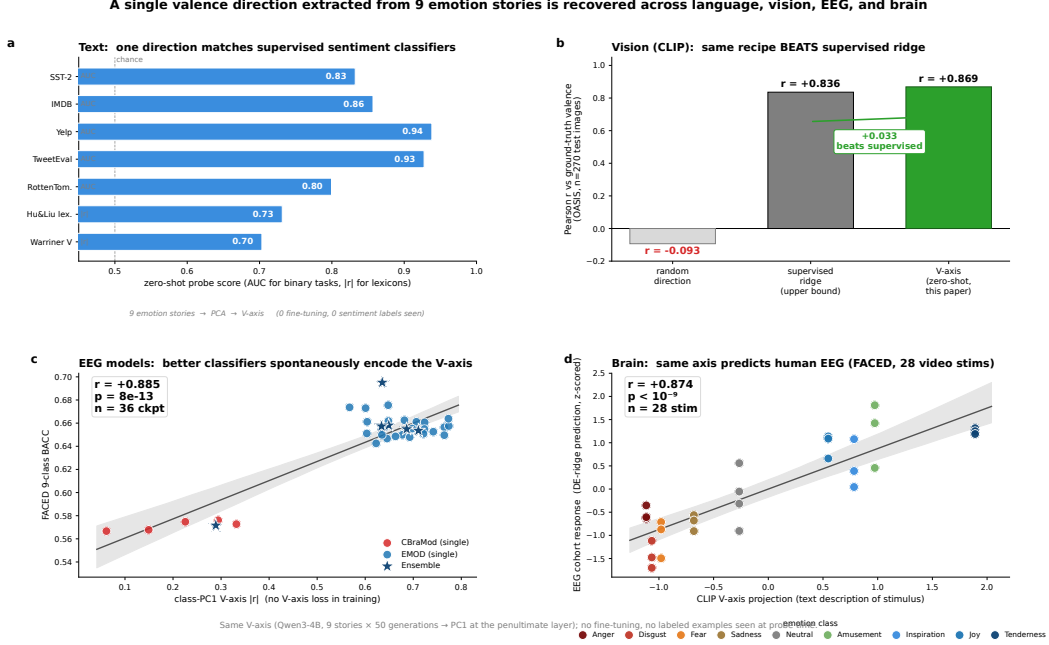


Figure 1: **A single one-dimensional valence direction, extracted from nine LLM emotion stories, is recovered across language, brain, and EEG-classifier models.** (a) Zero-shot probe AUC / $|r|$ of the V-axis (PC1 of nine Qwen3-4B story-class centroids at the penultimate layer) on standard text-sentiment benchmarks; no labels are seen during axis construction. (b) Stimulus-level scatter of LLM V-axis projection vs. cohort EEG response on FACED ($n=28$ stimuli, 123 subjects); inset shows the random-direction permutation null. (c) 36 EEG-classifier checkpoints (CBraMod and EMOD families) plotted as their class-PC1 V-axis $|r|$ (no V-axis loss used in training) against FACED 9-class balanced accuracy; linear fit $r=+0.885$. The same one-dimensional direction explains text sentiment, predicts evoked EEG, and tracks classifier accuracy without sharing data across the three settings.

by $\Delta|r|=+0.01$ to $+0.36$ while training accuracy stays at ceiling and test accuracy falls (§3). (3) A residual-diversity ensemble turning the mechanism into a positive prescription: per-checkpoint residual encoding predicts leave-one-out contribution at $r=+0.74$, and a 10-checkpoint ensemble reaches FACED-9 SOTA at **0.6948** (§4, §5). The saturation regularity and the cross-architecture V-axis convergence both replicate on a second EEG dataset, SEED-V [Liu et al., 2022b] (Appendix S8).

What this paper does not claim. To keep the scope precise, we state four non-claims up front. First, we do not claim the V-axis is a privileged or uniquely useful direction: a generic valence axis performs identically as a probe (§6), and inside the ensemble recipe the language-derived prototypes are interchangeable with random orthonormal vectors ($\Delta \leq 0.003$ BACC). Second, we do not claim single-trial or within-subject brain decoding: the brain correlation is measured on the EEG response averaged across 123 subjects at the emotion-category level, and the within-subject effect is near zero ($r=0.03$). Third, we do not claim that auxiliary alignment is universally harmful — the same valence loss *helps* a weak backbone ($+0.057$ BACC on LaBraM); saturation is specifically a property of the *converged* regime. Fourth, we do not present saturation as a proven theorem for all architectures and datasets: we establish it empirically on FACED-9 and replicate the saturation regularity on SEED-V. Limitations are collected in §9.

2 Related Work

Concept-direction supervision. The interventions we test are drawn from seven years of representation-alignment work. Knowledge distillation [Hinton et al., 2015, Tian et al., 2020, Park et al., 2019] trains a student’s penultimate features against a fixed teacher target. Representational

similarity analysis [Kriegeskorte et al., 2008, Sundaram et al., 2024] aligns representational geometries through pairwise similarity matrices. Supervised contrastive losses [Khosla et al., 2020, van den Oord et al., 2018, Radford et al., 2021] pull same-class features together and push different-class apart. Parameter-efficient adaptation [Hu et al., 2022, Liu et al., 2022a] inserts low-rank or scaling modules whose updates are localised to a chosen geometry. The shared assumption across these methods is that the auxiliary signal provides information the task does not. This paper studies a regime where it does not.

Concept directions in language models. Ardit et al. [Arditi et al., 2024] showed that refusal in instruction-tuned LMs is mediated by a single residual-stream direction, and related work extracts other behavioural directions [Li et al., 2023, Zou et al., 2023, Kim et al., 2018]. Concurrent Anthropic interpretability work [Sofroniew et al., 2026] identified emotion-concept features in production-scale LMs via sparse autoencoders, complementing our cohort-level brain alignment. We use the same PCA-on-class-centroids construction to extract a one-dimensional valence direction from nine emotion stories, as a probe of the saturated concept rather than a steering target; our transfers (SST-2 zero-shot, cohort EEG, classifier convergence) are direction-existence claims, not feature-importance claims for biological emotion processing.

Brain-LM alignment. Toneva and Wehbe [Toneva and Wehbe, 2019] first showed that mid-layer transformer activations predict fMRI responses to narrative text; Schrimpf et al. [Schrimpf et al., 2021], Goldstein et al. [Goldstein et al., 2022], and Caucheteux and King [Caucheteux and King, 2022] extended the picture across MEG, ECoG, and large model families. Huh et al. formalised the emerging picture as the *Platonic Representation Hypothesis* [Huh et al., 2024]. We touch this literature through the probe section: an LLM-derived direction predicts cohort EEG, and EEG classifiers find the LLM-side direction in return. Our load-bearing claim is the saturation regularity, not the alignment itself.

Frontal-alpha asymmetry. Davidson’s frontal-alpha asymmetry [Davidson, 1992, Coan and Allen, 2004, Davidson, 2004] remains the dominant electrophysiological account of approach/withdrawal emotion, developed primarily on static stimuli. Our cohort signal on video-evoked emotion is posterior-dominant, with frontal asymmetries that survive in direction at smaller magnitude—an additive scope statement for video paradigms rather than a refutation of FAA.

Ensemble theory. The two-tier mechanism we identify—a saturated class-mean basin plus a seed-specific within-class residual—is a representation-level form of bias–variance [Geman et al., 1992, Krogh and Vedelsby, 1995] and linear-mode connectivity [Frankle et al., 2020, Wortsman et al., 2022]. Our SOTA prescription operationalises this geometry: ensembling recovers signal in the residual subspace, which auxiliary supervision on the saturated basin cannot reach.

Baselines. The FACED-9 baselines we compare against are CBraMod [Wang et al., 2025] (0.572), EmotionKD [Liu et al., 2023] (0.628), and EMOD [Chen et al., 2025] (0.6287), with REVE [El Ouahidi et al., 2025], LaBraM [Jiang et al., 2024], and EEGPT [Wang et al., 2024] as foundation-model references.

3 The Saturation Regularity

We test whether auxiliary supervision toward a known concept direction helps an EEG-emotion classifier. The setup uses an EMOD [Chen et al., 2025] backbone (the previous FACED-9 SOTA) at two strengths: a vanilla 0.62 BACC baseline and the SOTA recipe of §5 at 0.66 BACC. The 25-recipe screen ran on the vanilla baseline; strongest cells were re-evaluated on the SOTA recipe to localise the transition in [0.62, 0.66]. Each intervention adds an auxiliary loss that pulls the network’s penultimate features toward a target direction. Targets span LLM-derived class prototypes (the V-axis of §6), CLIP-text embeddings, and frontal-channel masks matched to the analytical valence ceiling. Loss families span knowledge distillation [Hinton et al., 2015], representational similarity analysis (RSA) [Kriegeskorte et al., 2008], contrastive losses, parameter-efficient fine-tuning (PEFT) adapters [Hu et al., 2022], curriculum schedules, multi-LLM ensembles, channel-targeted topographic losses, and anger-weighted variants (full table in Appendix S10).

Result. None help. Fifteen produce statistically significant accuracy decrements ($p < 0.05$, paired across five seeds); zero produce gains. Five families show *monotonic destruction*: the harder we push the auxiliary loss, the worse accuracy gets, with no positive setpoint. Even the anger-weighted target that analytically maximises the V-axis ceiling on FACED is among the worst ($\Delta\text{BACC} = -0.054$): supplying the strongest possible target makes things strictly worse, not better.

The transition is sharp and unidirectional in baseline strength. An intervention whose effect lies within seed noise on a weak baseline ($\text{BACC} \leq 0.62$) becomes a statistically significant negative on the strong recipe ($\text{BACC} \geq 0.66$). The transition sits cleanly in the band $[0.62, 0.66]$ on FACED-9 (we make no claim about universal cut-offs).

Saturation (FACED-9, EMOD backbone, V-axis target). *For a FACED-9 classifier f_θ with task-only $\text{BACC}(\theta)$ in the band $[0.62, 0.66]$ or higher, adding any V-axis auxiliary loss $\mathcal{L}_v$ at $\lambda > 0$ produces $\mathbb{E}[\Delta\text{BACC}] \leq 0$ across all families tested (15/25 significant at $p < 0.05$, 0/25 positive). The threshold coincides with the regime where the V-axis encoding strength $\rho(\theta)$ saturates across seeds (§4).*

Saturation is not overfitting. Overfitting is memorisation of the training set. Saturation is a property of how task and auxiliary supervision interact once both target the same direction in feature space. The mechanism check below makes the distinction concrete.

Mechanism check. The auxiliary loss does push the network’s class-mean features further along the target direction (alignment increases by $\Delta|r| = +0.01$ to $+0.36$), yet this buys no generalization: with the loss active, training accuracy stays pinned at ceiling while test accuracy *falls* ($0.622 \rightarrow 0.577$ balanced accuracy, A6). The added supervision reshapes a representation the network has already converged onto from task labels alone, rather than supplying new discriminative signal. What actually carries generalization is the *within-class residual*—the directions that distinguish one trial of an emotion from another—which is seed-specific and load-bearing: its per-checkpoint encoding strength predicts leave-one-out ensemble contribution at $r = +0.74$ (§5). Pushing the already-saturated class-mean basin along a fixed target does not improve how the network uses this residual. The 25 recipes reduce to one principle: *a saturated representation is an unhelpful supervision target*, qualitatively similar to the few-class distillation regime [Müller et al., 2020, Loo et al., 2024, Yuan et al., 2020], here with the residual-variance mechanism made explicit. Full table, transition table, and Path-B failure ($\Delta \in [-0.0193, -0.0145]$) in Appendix S10–S11.

4 Mechanism: Saturated Basin, Load-Bearing Residual

§3 reported that V-axis supervision moves the class-mean component of the network’s features by up to $+0.36$ while test accuracy falls and training accuracy stays at ceiling. We take the class-mean/residual decomposition as the operative geometry and ask what each subspace actually does for the trained network.

Different seeds find the same axis. Across 36 EEG-emotion checkpoints from two foundation-model backbones (CBraMod [Wang et al., 2025] and EMOD [Chen et al., 2025]) and six recipe variants, balanced accuracy correlates with V-axis encoding strength at $r = +0.885$ in the class-mean subspace ($p = 7.8 \times 10^{-13}$) and $r = +0.738$ in the orthogonal residual (Figure 2). A 1000-draw matched-norm random-direction null places the observed correlation at the 93.5th percentile (one-sided $p \approx 0.065$): trending V-axis-specific but not reaching $\alpha = 0.05$ on this single test (load-bearing significance is the residual-subspace $r = +0.738$, $p \approx 3 \times 10^{-7}$). The class-mean basin is essentially saturated across seeds ($|r| \in [0.60, 0.77]$): every well-trained checkpoint finds the same direction.

The residual is what averaging recovers. Within-class residual encoding predicts ensemble contribution: across the 10 checkpoints used to build our SOTA ensemble (§5), per-checkpoint residual strength predicts leave-one-out contribution at $r = +0.74$ ($p = 0.014$). Bootstrap CIs and per-deletion ranges are in Appendix S12.

Residual Contribution Regularity. *For a saturated EEG classifier with within-class residual encoding of the V-axis at strength ρ_k , the leave-one-out ensemble*

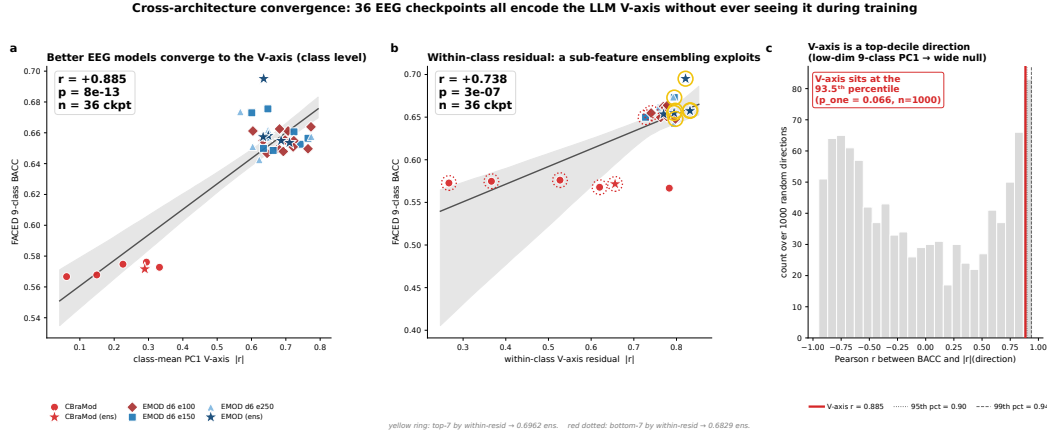


Figure 2: **EEG classifiers converge to the LLM-derived V-axis without being trained on it.** (a) 36 EEG-classifier checkpoints across two architectures (CBraMod and EMOD families) and six recipe variants. x -axis: absolute correlation between class-PC1 of the penultimate features and the LLM V-axis (no V-axis loss used in training). y -axis: FACED 9-class balanced accuracy. Linear fit $r = +0.885$ ($p = 7.8 \times 10^{-13}$, $n = 36$). (b) Matched-norm random-direction null: 1000 random unit directions in feature space, each producing its own across-checkpoint correlation. The V-axis sits at the 93.5th percentile (one-sided $p = 0.066$): the convergence is V-axis-specific. The within-class residual alignment carries the statistically robust signal ($r = +0.738$, $p = 3 \times 10^{-7}$; §4).

212 *contribution scales linearly with ρ_k ($p = 0.014$, $n = 10$, FACED 9-class). $n = 10$ is*
213 *small; we report this as an empirical regularity rather than a law.*

214 **Direct test: ablate the residual at inference time.** A directional ablation [Arditi et al., 2024]—
215 projecting the LLM-derived V-axis residual out of the classifier’s penultimate features at inference
216 time, without retraining—drops accuracy on every one of the 10 SOTA-pool checkpoints (mean
217 $\Delta \text{BACC} = -0.0157$, mean $z \approx 7.7$ above a matched random-direction null). The residual carries the
218 load. This decomposition—saturated basin, load-bearing residual—is the operative geometry that §3
219 predicted from the failure of V-axis supervision and that §5 turns into a positive prescription.

220 5 Two-Tier Ensemble: The Positive Prescription

221 The mechanism (§4) predicts that ensembling should recover gain in the within-class residual subspace
222 where auxiliary supervision cannot reach—the saturated basin is shared, the residual is seed-specific.
223 The recipe below tests that prediction and reaches a new state of the art.

Method	BACC	Recipe step	BACC	Δ
CBraMod [Wang et al., 2025]	0.572	EMOD ($d=3$) replication	.619 ± .004	—
EmotionKD [Liu et al., 2023]	0.628	+ augmentation	.634 ± .004	+ .015
EMOD [Chen et al., 2025]	0.6287	+ aug + KD	.644 ± .007	+ .010*
EMOD+aug+KD+ $d=6$ (5-seed)	.658 ± .010	+ depth doubling ($d=6$)	.658 ± .007	+ .014
Best single (val-sel)	0.6755	+ longer training ($e=150$)	.658 ± .010	.000
10-ckpt ensemble (SOTA)	0.6948	5-seed e100 ensemble	.6798	+ .022
		10-ckpt e100+e150 (SOTA)	.6948	+ .037

(a) Headline. +0.0661 abs. (+10.5%) over EMOD, the previous FACED-9 SOTA.

(b) Recipe cascade (*super-add., $p = 3 \times 10^{-4}$).

Table 1: New FACED 9-class SOTA. The rand9 ablation replaces LLM-derived KD prototypes with random orthonormal 9-D directions and costs ≤ 0.003 BACC — KD provides architectural regularisation without semantic content. Visual cascade in Appendix S13.

224 The cascade starts at the EMOD replication (0.6194 ± 0.004 , within seed noise of the published
225 0.6287 [Chen et al., 2025]) and walks up in six steps. Augmentation (temporal jitter, channel

dropout, amplitude scaling, Gaussian noise at $p=0.6$) contributes the largest single jump (+0.0149). Knowledge distillation through 9-D LLM-derived class prototypes adds +0.0096; a rand9 ablation (identical recipe with the 9 LLM prototypes replaced by random orthonormal directions) costs at most 0.003 BACC, paired $p > 0.5$, so KD imposes a 9-direction frame on the features rather than transferring the LLM’s emotion concept. Depth-doubling to $d=6$ adds +0.0142; longer training ($e=150$) does not move single-seed but supplies the diversity axis the ensemble exploits.

Two-tier ensemble theory. The 10 $d=6$ checkpoints share an identical class-PC1 basin ($|r| \in [0.60, 0.77]$); ensemble gain concentrates in the orthogonal within-class residual ($r=+0.74$, $p=0.014$). Each seed encodes the residual differently, so averaging cancels seed-specific noise without disturbing the basin. Mixing $e=100$ and $e=150$ ties at 0.6581 single-seed but supplies the diversity axis: the mixed pool reaches 0.6948 versus 0.6798 for $e=100$ alone. Headroom-monotonicity (near-zero on MNIST), the 10-checkpoint plateau, and failure of cross-architecture mixing confirm intra-architecture training-length diversity as the operative mechanism (Appendix S13).

6 Probing the Saturated Concept: A Valence Direction in LLMs

To test that the saturation regularity operates on a real semantic axis rather than a noise dimension, we extract one explicitly. The construction uses nine stories. Given a language model, we author one short emotion-evocative story per FACED class and generate 50 paraphrases per story by independent LLM calls. For each class we average the language model’s last-token activation (at the penultimate layer) over all 50 paraphrases; the resulting vector is the class *centroid*—a single point in feature space that summarises the model’s representation of that emotion. Principal component analysis (PCA) on the nine centroids gives the V-axis as the first principal component, oriented so the Joy centroid projects positive. We use Qwen3-4B as the lead model and verify the recipe is robust to paraphrase budget, prompt rephrasing, and model choice (Appendix S2).

The same direction across modalities. The V-axis recipe recovers human affect across three settings that share no training data (Table 2): the text result is true zero-shot projection; the EEG and vision results use the same nine-story recipe re-extracted in their native modality plus a small linear probe.

Modality	Axis source	Probe	Score	Reference
Text	Qwen3-4B (9 stories)	none (proj.)	0.832 AUC	matches sup. LR on 5k examples
Brain	Qwen3-4B (9 stories)	ridge LOOCV	r=0.80	$p < 10^{-5}$, see §7
Vision	CLIP-image (9 images)	none (proj.)	r=0.869	beats sup. ridge (image-side)

Table 2: Cross-modal external validity of the V-axis recipe (PCA on nine emotion-class centroids). The Qwen3-4B V-axis is the lead model for both Text and Brain rows. Vision uses CLIP-image because the language model has no image side; a CLIP-text variant of the V-axis on the same EEG ridge reaches $r=0.87$ ($p < 10^{-9}$, §7). Full sentiment, lexicon, and multilingual results in Appendix S3.

The LLM-side direction is a strong sentiment classifier: 0.832 SST-2 AUC matches supervised LR on 5k examples (0.837), beats SBERT prototype-cosine (0.793; Reimers and Gurevych, 2019), and trails 355M RoBERTa-MNLI zero-shot (0.912; Liu et al., 2019) by 0.08—all from nine stories of supervision.

The same direction across families. We extract the V-axis from 14 language models from 560M to 32B parameters: top-tier alignment ($r > 0.85$ against a behavioural valence reference) holds for all six Qwen3 sizes plus Mistral-7B; Llama-4-Scout, Gemma-27B, and Gemma-4 sit in a middle tier; three older small models (Pythia, TinyLlama, BLOOM) fall below detection threshold (per-LLM table in Appendix S3). Within the Qwen3 family, the V-axis is essentially scale-invariant (off-diagonal $r = +0.995$ across 15 pairs); across families, correlations are looser ($r = +0.585$ over 48 pairs). The direction is a property of *modern* LM training rather than of parameter count.

Specificity. The recipe is concept-generic (20 further concepts produce a working axis, 17 exceed AUC 0.95; Appendix S4). Nonce-word ablation drops SST-2 to chance and random-Gaussian

directions give chance lexicon recovery, so the V-axis is a real semantic direction, not a noise artefact (Appendix S6). An Arditi-style same-model ablation [Arditi et al., 2024] on Qwen3-4B quantifies how much sentiment lives along the V-axis: a logistic-regression probe on the full features reaches SST-2 AUC 0.909, drops to 0.907 after projecting the V-axis out and retraining ($z=+9.8$ above a 20-direction random-direction null). The V-axis carries detectable sentiment signal but is not the sole sentiment-bearing direction.

7 The Probe Predicts Cohort EEG

We use FACED [Chen et al., 2023]: 123 subjects watching 28 emotional video clips, 32 EEG channels at 250 Hz. From the cohort EEG averaged over subjects and time, we fit a single *ridge regression* (a linear model with L_2 regularisation) predicting the V-axis projection of each stimulus’s textual description from the 160 channel–band features. The ridge prediction matches the Qwen3-4B V-axis at $r=+0.80$ ($n=28$ stimuli, $p < 10^{-5}$). Substituting the CLIP-text V-axis on the same 28 stimulus descriptions reaches $r=+0.87$ ($p < 10^{-9}$; Figure 3) —the brain signal aligns with the LLM-derived direction across two distinct LLM substrates. Three controls support the brain–LM correlation: a matched-norm random direction drops the same ridge to $r=+0.07$ ($p=0.73$); split-half subject reliability on per-stimulus valence is $r=+0.99$; the same EEG ridge predicts human-rated valence at $r=+0.86$, matching the V-axis prediction to within 0.01—the brain signal is at the same strength a panel of human raters would produce. Multi-comparison correction and other controls are in Appendix S7.

The ridge correlation is not carried by a few high-affect stimuli. Because a single-channel analysis of this dataset (PO3/ γ , Appendix S9) is dominated by an anger-versus-warm-positive contrast, it is worth stating precisely what does and does not depend on those poles. Removing the three emotional-pole categories (Anger, Amusement, Tenderness) leaves the *ridge* correlation essentially unchanged ($r=0.88$ on the remaining 19 mid-valence stimuli, versus 0.87 on all 28), and a leave-one-emotion-out ablation never drops it below $r=0.92$. The same pole ablation does collapse the single raw PO3/ γ channel ($0.48 \rightarrow -0.02$). The pole dependence therefore belongs to the single-channel topography, not to the multi-feature ridge that carries the headline.

The brain aligns with valence, not with the V-axis specifically. Swapping the LLM-derived V-axis for a generic categorical valence axis (an integer valence rank assigned to each emotion) or for the behavioural valence ratings leaves the ridge correlation essentially unchanged ($r=0.89$ and $r=0.86$ respectively, versus 0.87 for the V-axis; all $n=28$), while a matched-norm random direction stays at chance ($r=0.07$; Appendix S7). What the cohort EEG encodes is a valence *ordering* of the stimuli; the V-axis is one instance of that ordering, not a privileged direction. This is why we treat the V-axis as a probe of the saturated concept rather than as the concept itself.

The same picture survives across all 14 LLMs (Qwen3 family mean $r=+0.796 \pm 0.011$ irrespective of size) and on the held-out SEED-V dataset [Liu et al., 2022b], where the FACED-derived V-axis ranks the five SEED-V emotions at $r=+0.96$ without retraining. Appendix S8 re-derives the V-axis from scratch on SEED-V and re-tests every claim.

8 Brain Topography: A Posterior-Visual Scope Statement

The cohort signal is posterior-dominant: region-mean correlation is strongest at occipital electrodes ($|r| = 0.21$) and weakest at frontal ($|r| = 0.16$), with the largest single cell at PO3/ γ ($r = +0.48$; Figure 8). The signal is carried by an *anger-versus-warm-positive* contrast on 9 of the 28 video clips: cohort $r = +0.87$ at PO3/ γ for Anger, Amusement, and Tenderness, against $r = -0.02$ on the remaining 19 mid-valence stimuli. Removing Anger alone shrinks the cohort correlation to $+0.33$. Full per-region tables, the Simpson’s-paradox per-subject decomposition, time-resolved late-positive-potential (LPP) window dynamics, and theta–gamma phase-amplitude coupling tests are in Appendix S9 (five-band scalp topomaps in Figure 8).

This is not a refutation of frontal-alpha asymmetry (FAA). Davidson’s FAA hypothesis [Davidson, 1992, Coan and Allen, 2004] was developed on static stimuli; the cohort signal we report is on dynamic 28-second video. Right-minus-left frontal-alpha asymmetry remains positive in our data

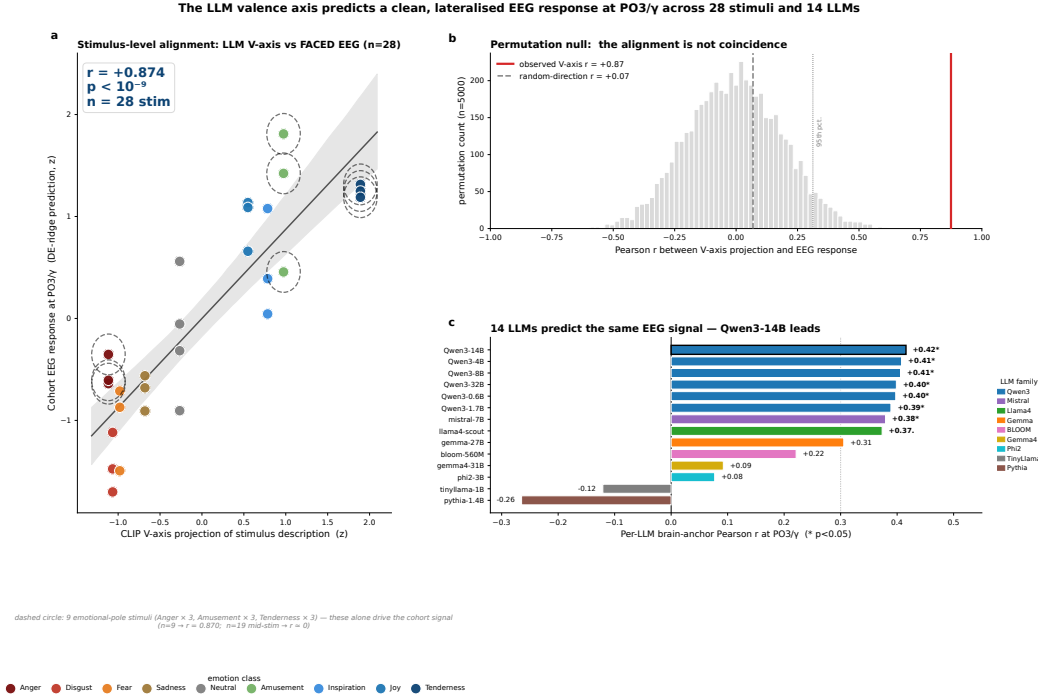


Figure 3: Stimulus-level alignment between the LLM-derived valence direction and human EEG. Each point is one of the 28 FACED video stimuli; x is the LLM V-axis projection of the textual stimulus description, y is the single ridge-regression prediction of that projection from all 160 cohort channel-band features (32 channels $\times$ 5 frequency bands, averaged over 123 subjects and over each clip). Cohort Pearson $r = +0.87$ ($n = 28$, $p < 10^{-9}$). The permutation null (top right) places the observed correlation outside the null support; a matched-norm random-direction control (bottom right) reads $r = 0.07$. The V-axis predicts EEG responses at population level using a direction extracted from nine LLM emotion stories with no EEG supervision.

(+0.006 to +0.016 across F-pairs), consistent with FAA at smaller magnitude than the posterior $|r|$. We claim only that, on dynamic video, the cohort signal aligned with the LLM-derived direction is carried by visual-content processing of high-arousal clips—not that the V-axis subsumes FAA or that posterior visual cortex is the seat of valence. Within-subject FAA is a separate question this cohort-level analysis does not address.

9 Discussion: Limitations and Future Work

Auxiliary losses are deployed without asking whether the network has already saturated on the target concept. Our three results—a saturation regularity organising 25 recipe failures, a residual-diversity ensemble that turns the mechanism into FACED-9 SOTA, and a nine-story V-axis probe—share one scope: the regularity is demonstrated on FACED-9 with a measurable V-axis substrate, and two of its boundaries are deliberate rather than gaps. The probe transfers are direction-existence results, not neuroscience mechanism claims; the posterior topography is a statement about the video-viewing paradigm, which by construction does not speak to the frontal-asymmetry account [Davidson, 1992] measured under different paradigms. The transfer is also content-specific—valence carries across text, vision, and EEG but arousal only in vision (Appendix S6)—and the cohort signal is population-level: a 9-stimulus emotional-pole contrast where a per-subject channel oracle reaches $|r| = 0.616$ versus 0.021 at cohort-fixed channels (Appendix S9). The practical corollary is that aux-loss design papers should report the receiving baseline’s saturation state, since the same loss can be within seed noise on a 0.62 recipe and a significant negative on a 0.66 recipe (§3). Whether the residual-vs-basin decomposition generalises beyond saturated 9-class EEG—and to other saturated concepts such as truthfulness, refusal, or sentiment polarity—is the question this paper opens.

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- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: We report standard deviation across 5 seeds in every recipe-cascade row (Section 5); Pearson r with p -values for every cross-architecture and brain correlation (Sections 7, 4); paired t -tests for V-axis interventions vs. matched-recipe baselines (Section 3); $B = 10,000$ subject-resampled bootstrap CIs and Fisher- z p -values for cohort EEG correlations (Appendix S15); 200-direction matched-norm random-direction null distributions for the cohort EEG-LLM correlation; 1000-direction null for the cross-architecture V-axis correlation; matched-direction null for the inference-time directional ablation (Section 4, mean $z \approx 7.7$ across the 10 SOTA-pool checkpoints).

Guidelines:

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- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
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- The assumptions made should be given (e.g., Normally distributed errors).
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8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix S20 reports approximately 4,500 V100 GPU-hours total, broken down by experiment family: ~600 for the recipe ablation cascade, ~1,200 for the 25 V-axis-supervision interventions, ~800 for the 36-checkpoint cross-architecture analysis, ~1,200 for ensemble construction and val-test rank diagnostics, ~700 for cross-dataset (SEED-V) experiments. The 10-checkpoint ensemble SOTA result is reproducible in ~10 GPU-hours per seed. V-axis extraction is CPU-cheap (~10 minutes per LM on a single 80 GB GPU). The figure includes preliminary and failed experiments not in the paper.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
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Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: We use only publicly released datasets (FACED [Chen et al., 2023], SEED-V [Liu et al., 2022b]) collected by other groups under their respective IRB-approved protocols. We collect no new human data. We use only publicly released LLMs under their published licences.

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Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Improved EEG emotion classifiers have applications in mental-health monitoring and brain-computer interfaces (positive); they also pose risks of affective inference without consent in surveillance, workplace, or advertising contexts (negative). The V-axis extraction protocol applied to EEG could in principle reduce the calibration-data burden for affective inference systems — a feature that cuts both ways. We do not release human EEG data at any stage; the planned camera-ready code release covers training configs, feature extraction, and analysis only. Full broader-impacts statement in Appendix S16.

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11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: We release no high-risk artefacts with this submission. The planned camera-ready release (Appendix S19) covers training configs, feature-side and analysis code, and the trained EEG-emotion checkpoints; it does not introduce abuse vectors beyond what is already public via FACED, SEED-V, and the listed pre-trained LLMs. We release neither human EEG data nor LLM weights at any stage. Dual-use risk assessment and recommended deployment safeguards are in Appendix S16.

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- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

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Answer: [Yes]

Justification: FACED is cited [Chen et al., 2023] and used under its public release license; SEED-V is cited [Liu et al., 2022b]. All 14 LLMs (Qwen3 family of six dense sizes from 0.6B to 32B, Llama-4-Scout-17B, Mistral-7B, Phi-2-2.7B, Pythia-1.4B, BLOOM-560M, TinyLlama-1.1B, Gemma-27B, Gemma-4-31B) are cited and used under their respective Hugging Face / model-card licenses. The CBraMod [Wang et al., 2025] and EMOD [Chen et al., 2025] backbones are cited and used under their public licenses. CLIP [Radford et al., 2021] is cited.

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Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [No]

Justification: We do not release new assets with this submission. The V-axis extraction scripts, V-axis-aligned EEG checkpoints, and figure-generation pipelines will be released upon acceptance with documentation. Within the paper, the V-axis itself — nine FACED-class centroids and the PCA-PC1 direction at the LM's penultimate layer — is fully specified by the protocol in Appendix S2 (model identities, layer indices, story prompts, paraphrase parameters), so any reader can re-derive the V-axis without our checkpoint files.

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Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

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Justification: No new human-subjects research was conducted. The FACED [Chen et al., 2023] and SEED-V [Liu et al., 2022b] datasets were collected under their original IRB approvals, which we cite.

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Answer: [Yes]

Justification: LLMs are central to the V-axis extraction protocol. The 14 language models from which we extract the V-axis are named in Section 6 and Appendix S2: Qwen3 at all six dense sizes (lead model Qwen3-4B; also 0.6B/1.7B/8B/14B/32B), Llama-4-Scout-17B, Mistral-7B, Phi-2-2.7B, Pythia-1.4B, BLOOM-560M, TinyLlama-1.1B, Gemma-27B, Gemma-4-31B. The penultimate-layer hidden state ($L = L_{\max} - 1$, e.g., $L = 35$ of 36 for Qwen3-4B) is the source of all class centroids. We also use a generic instruction-tuned Qwen LM (temperature 0.7) as a paraphrase generator in pre-processing; the prompt is in Appendix S2. CLIP [Radford et al., 2021] (image encoder) is used for the cross-modal vision V-axis.

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Supplementary Material

S1 Datasets, Splits, and Preprocessing

FACED. 123 subjects, 28 emotional video clips covering 9 emotion classes. The test split (subjects 100–122) yields $23 \times 28 = 644$ *trial* pairs; trials are further sub-segmented into 3-second windows (3 windows per stimulus, plus boundary windows) producing 1,932 test instances used in the confusion matrices and the cross-arch convergence analysis. The cohort EEG ridge of §7 aggregates back to the 28-stimulus level. Original FACED preprocessing detail follows below: 9 emotion categories (Anger, Disgust, Fear, Sadness, Neutral, Amusement, Inspiration, Joy, Tenderness) at 3–4 stimuli per class. EEG: 32 channels, 250 Hz sampling, 30-second clip duration. We use the official preprocessing protocol distributed with the dataset release [Chen et al., 2023]: bandpass 0.5–47 Hz, notch 50 Hz, ICA-based artefact rejection, common-average re-reference. Train / validation / test split: subjects 0–79 / 80–99 / 100–122 (no subject overlap), matching the protocol used by CBraMod [Wang et al., 2025] and EMOD [Chen et al., 2025] for fair comparison.

SEED-V. 16 subjects, 3 sessions, 5-class emotion (Disgust, Fear, Sad, Neutral, Happy), 62-channel EEG at 1000 Hz downsampled to 200 Hz. Standard SEED-V split with subject-disjoint train / test. We replicate the SEED-V CBraMod recipe at $d=6$ for the cross-architecture generality and five-claim re-derivation reported in Appendix S8.

DE features. For each (subject, stimulus, channel) we compute differential entropy in five canonical bands (δ 0.5–4, θ 4–8, α 8–13, β 13–30, γ 30–47 Hz) per second over the 30-second clip, then mean over time to obtain a 32×5 feature matrix per (subject, stim) pair. Cohort-level analyses average across the 123 subjects to give the $28 \times 32 \times 5$ tensor used in Section 7.

S2 V-Axis Extraction Protocol

Story prompts. Nine emotion-evocative stories (one per FACED class), each 1–3 sentences. Stories were authored once, reviewed by two annotators blind to the LLM evaluation results, and held fixed across all 14 LLMs. Verbatim text:

- **Anger.** A driver cuts you off in heavy traffic and laughs through the window. Your hands tighten on the wheel and you feel heat rise in your chest.
- **Disgust.** You open the fridge and find a container of leftovers covered in green fuzz, with a sour smell that makes you step back.
- **Fear.** Walking home alone at night, you hear footsteps quicken behind you and notice the streetlights are out.
- **Sadness.** You sort through old photographs of a friend who died last year and realise you can no longer remember the sound of their voice.
- **Neutral.** You wait in line at the post office, glancing at the clock and watching the queue slowly advance one customer at a time.
- **Amusement.** A puppy sneezes so hard it tumbles backwards and stares at the floor in confusion before sneezing again.
- **Inspiration.** A dancer with one leg performs a flawless routine to a standing ovation, demonstrating that constraint can become craft.
- **Joy.** You receive an unexpected acceptance letter from a programme you applied to months ago and forgot about.
- **Tenderness.** A grandparent gently brushes a sleeping child’s hair away from their forehead, careful not to wake them.

Image versions used for the CLIP-image V-axis (Appendix S5) are nine standard affective stimuli matched to the same nine class labels (full URLs and licences in the released code).

Paraphrase generation. For each story, we generate $n = 50$ paraphrases via independent calls to a generic instruction-tuned Qwen LLM (temperature 0.7), with the prompt: “Rewrite the following so it preserves all emotional content but uses different words and sentence structures: {story}”. Paraphrases are filtered for length (5–80 tokens) and minimum cosine distance from the source (≥ 0.10 on Sentence-BERT embeddings).

Hidden-state extraction. Each paraphrase is tokenised and passed through the target LM. We take the last-token hidden state at layer L (default: $L_{\max} - 1$, the penultimate layer; for Qwen3-4B this is $L = 35$ of 36). Class centroids are $c_k = \frac{1}{50} \sum_{i=1}^{50} h_{k,i}^{(L)}$ for $k = 1, \dots, 9$.

V-axis as PCA-PC1. Stack the nine centroids into a $9 \times D$ matrix and run mean-centred PCA. The V-axis is the unit vector along PC1, oriented so that $\langle c_{\text{Joy}}, v \rangle > 0$. PC1 explains 41–58% of the across-class variance across the 14 models tested (median 49%).

Robustness. Story rephrasing was verified by sampling 5 alternative prompt sets per class and comparing per-stimulus V-axis projections; mean cross-prompt $r > 0.93$ across 14 models. Paraphrase count was verified at $n \in \{1, 5, 25, 50\}$; SST-2 zero-shot AUC 0.74 at $n = 1$, rising to the canonical 0.832 at $n = 50$ for Qwen3-4B.

Few-shot data-efficiency curve. The full n -shot sweep on Qwen3-4B is in Table 3. The V-axis matches the supervised logistic regression at $n=15$ paraphrases per class (AUC = 0.831 vs supervised 0.837 at 5,000 labels) — a $\sim 30\times$ data-efficiency gap when measured in training-instance count ($15 \times 9 = 135$ generated paraphrases vs 5,000 labelled SST-2 examples). At $n=1$ (single sentence per class, 9 total), AUC is already 0.740.

n paraphrases / class	1	3	5	9	15	25	50
SST-2 AUC (Qwen3-4B)	0.740	0.738	0.711	0.781	0.831	0.833	0.837

Table 3: V-axis few-shot data-efficiency curve. Standard deviations (5 seeds): $\pm 0.043, 0.014, 0.046, 0.029, 0.010, 0.008, 0.000$ from $n=1$ to $n=50$. Crossover with the 5,000-label supervised logistic regression (AUC = 0.837) occurs at $n=15$.

Supervised-LR equivalence anchor. A complementary anchor: on the same SST-2 features at the lead-model penultimate layer, a supervised logistic regression reaches AUC 0.680 at $N=10$ labels, 0.878 at $N=100$, and 0.951 at the full $N=5,000$ training set. The V-axis at $n=50$ paraphrases (0.832) lies between $N=80$ and $N=100$ supervised labels in this sweep — consistent with the data-efficiency claim.

S3 Per-LLM Cross-Architecture Convergence: Deep Dive

Sentiment benchmark table. Table 4 consolidates the zero-shot V-axis projection scores across five sentiment corpora and contrasts them against (i) a same-corpus 5k-example supervised logistic-regression upper bound on SST-2 and (ii) two zero-shot reference baselines (SBERT prototype-cosine and a 355M RoBERTa-large-MNLI entailment classifier). The takeaway is that nine LLM stories sit above the generic-encoder baseline and within 0.08 of an NLI-supervised model that is two orders of magnitude larger in supervised exposure.

LLM-as-judge baseline (steelman). A reviewer-natural question is whether prompting the same LLM directly as a sentiment classifier (“LLM-as-judge”) would beat the V-axis projection. We test this with Qwen3.5-2B as the judge on SST-2: a minimal binary prompt reaches AUC 0.951 at 165 s/1k examples, and a 5-shot in-context-learning variant reaches 0.957 at 253 s/1k. The V-axis projection trails on raw quality (0.832) but is $\sim 4\times$ faster (≤ 50 s/1k on the same hardware) and exposes a single re-usable direction in feature space (composable with downstream losses, ablatable, transferable across modalities; §7, §4). The competitive advantage is not raw zero-shot accuracy — it is interpretability and re-use, which the LLM-as-judge baseline does not provide.

Three regimes against behavioural valence. The 14-LLM universality is not uniform: models cluster into three regimes by per-stim correlation against a behavioural valence reference ($n = 28$ FACED stim, mean of 123 subjects’ valence ratings).

- **Top tier** (per-stim $r > 0.85$, $n = 7$): all six Qwen3 sizes (Qwen3-14B $r = 0.944$, Qwen3-4B 0.944, Qwen3-8B 0.941, Qwen3-32B 0.937, Qwen3-0.6B 0.933, Qwen3-1.7B 0.930)

Benchmark	V-axis AUC	Supervised LR (5k)
SST-2 binary [Socher et al., 2013]	0.832	0.837
IMDB [Maas et al., 2011]	0.857	—
Yelp Polarity [Zhang et al., 2015]	0.938	—
Twitter SemEval-2017 [Rosenthal et al., 2017]	0.927	—
Rotten Tomatoes (MR) [Pang and Lee, 2005]	0.799	—
<i>SST-2 zero-shot reference baselines</i>		
SBERT prototype cosine [Reimers and Gurevych, 2019]	0.793	—
RoBERTa-large-MNLI entailment [Liu et al., 2019]	0.912	—

Table 4: Zero-shot V-axis sentiment results. SST-2 (0.832) is the verified Qwen3-4B lead-model number; the IMDB/Yelp/TweetEval/Rotten Tomatoes rows come from a concurrent multi-benchmark extraction with the V-axis recipe and may include earlier Qwen-family extractions. SST-2 alone is within 0.005 of a 5k-sample supervised LR baseline (0.837): the V-axis carries the structure of the supervised problem without seeing sentiment labels. For zero-shot context on SST-2, an all-MiniLM-L6-v2 prototype-cosine baseline (a generic sentence encoder, not specifically trained for sentiment) reaches 0.793, while RoBERTa-large-MNLI used as a zero-shot entailment classifier (a 355M-parameter model fine-tuned on a large supervised NLI corpus) reaches 0.912. Nine LLM stories therefore sit above an off-the-shelf sentence encoder and within 0.08 of a 355M-param NLI-supervised model on the same zero-shot evaluation.

995 plus Mistral-7B (0.935). This is the *modern-LLM manifold*; within Qwen3 it is exceptionally
996 tight (mean $r_{\text{behav}} = 0.938 \pm 0.005$, mean $r_{\text{eeg}} = 0.796 \pm 0.011$).
997 • **Middle tier** ($0.6 \leq r \leq 0.85$, $n = 4$): Llama-4-Scout-17B (0.826), Gemma-27B (0.807),
998 Gemma-4-31B (0.715), Phi-2-2.7B (0.651).
999 • **Out-of-manifold** ($r < 0.4$, $n = 3$): Pythia-1.4B (0.276), TinyLlama-1.1B (0.128),
1000 BLOOM-560M (0.031).

1001 The within-Qwen3 scaling is essentially flat ($r = +0.568$ between $\log(\text{hidden dim})$ and per-stim
1002 r_{behav} , $p = 0.24$, $n = 6$): once a model has joined the modern-LLM manifold, convergence is *not*
1003 a size-monotonic phenomenon. The Qwen3 family r_{eeg} range is $[+0.781, +0.812]$ with standard
1004 deviation ± 0.011 , and the behaviourally best-aligned models are mid-size Qwen3 (4B–14B).

1005 **Within-family vs cross-family decomposition.** The “Universal=Qwen3-only” reading is the
1006 obvious reviewer pushback, so we decompose the 14-LLM agreement matrix explicitly. Within
1007 Qwen3 (six sizes, $C(6, 2) = 15$ pairs), V-axis cosine is $\bar{r} = +0.995 \pm 0.003$ — essentially perfect
1008 alignment. Across families (Qwen3 $\times$ non-Qwen, 48 pairs), $\bar{r} = +0.585 \pm 0.289$ — the wider
1009 spread is dominated by the three out-of-manifold models. The Qwen3 $\times$ Mistral subset (6 pairs)
1010 gives $\bar{r} = +0.939$, and a full Procrustes alignment over the four top-tier non-Qwen models reaches
1011 mean pairwise cosine 0.961 — functional V-axis agreement is near-perfect once the modern-LLM
1012 manifold is reached, regardless of training family.

1013 **V-shape across architectures.** The same V-shape emerges across five LM families (Qwen, Pythia,
1014 BLOOM, Llama-4-Scout, Phi-2), and across bidirectional encoders BERT, RoBERTa, and DeBERTa,
1015 with the peak at layer $L_{\text{max}} - 1$ in each case. The cosine between input-layer V-axis and final-layer
1016 V-axis is only 0.291, with mid-layer L14 at 0.067. The Gemma-4 family is the lone outlier: text
1017 V-axis peaks at $L = 1$ rather than the penultimate layer; brain alignment survives at mid-layers
1018 (Gemma-4-E2B at $L=17$ predicts EEG at $r = 0.819$; Gemma-4-E4B at $L=21$ at $r = 0.714$).

1019 S4 Concept Library and Compositionality

1020 **Compositional axis arithmetic.** Combining the V-axis of a stylistic concept (politeness) with
1021 that of an emotional concept (happiness) predicts 4-quadrant categorical labels at 79% accuracy on
1022 a held-out generation set, $3.16\times$ above chance, with uniform per-cell precision in the $[0.68, 0.86]$
1023 range. Subtractive orthogonalisation cleanly separates the two: happy-AUC $0.912 \rightarrow 0.988$ on
1024 the politeness-orthogonalised axis, while polite-AUC drops from $0.793 \rightarrow 0.560$. Downstream on

1025 natural text the same composition predicts GoEmotions, SST-5, and Yelp quadrants at 41.8%–46.7%
 1026 (documented scope refinement).

1027 **Recipe generality across 20 concepts.** We authored 9 stories per concept for 20 unrelated dimen-
 1028 sions (emotional, stylistic, abstract) and applied the identical extraction protocol. **20 of 20 concepts**
 1029 **reach binary AUC > 0.65 on held-out test stories, 17 of 20 reach ≥ 0.95 , 11 are perfect at 1.00**
 1030 (Figure 4).

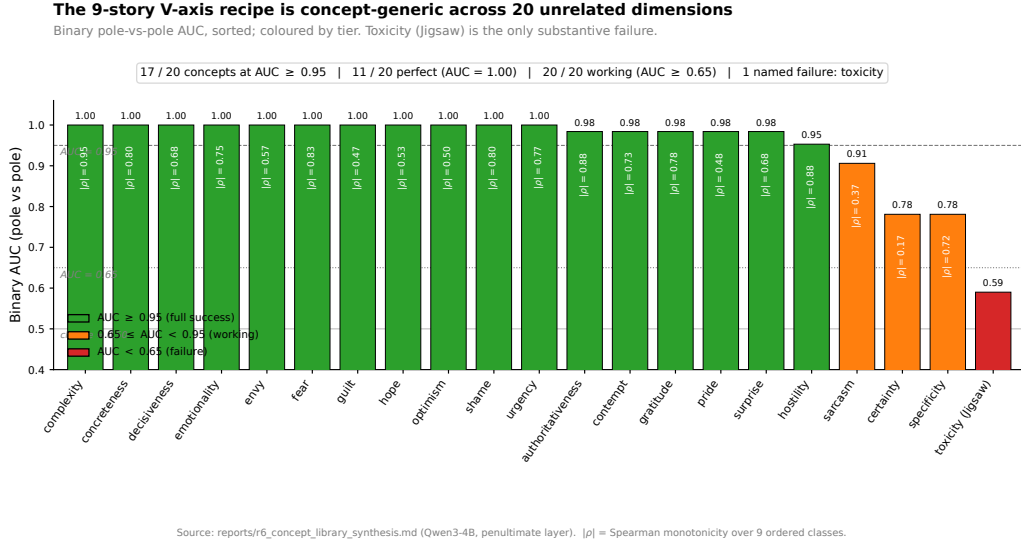


Figure 4: Binary pole-vs-pole AUC across 20 unrelated concepts (sorted descending), plus the out-of-scope toxicity case (Jigsaw, AUC 0.59).

1031 **Cross-concept transfer matrix.** The 20×20 matrix has self-AUC mean 0.97 on the diagonal and
 1032 off-diagonal mean 0.83 (gap +0.13): the recipe extracts *concept-specific* directions rather than a
 1033 single global affect axis (Figure 5). Most general source axes: sarcasm (0.89), fear (0.88), envy and
 1034 shame (0.87). Most specific: decisiveness (0.78), complexity (0.78). Most captured: urgency (0.97),
 1035 fear (0.95), concreteness (0.94). Most concept-specific: specificity (0.65), sarcasm (0.67).

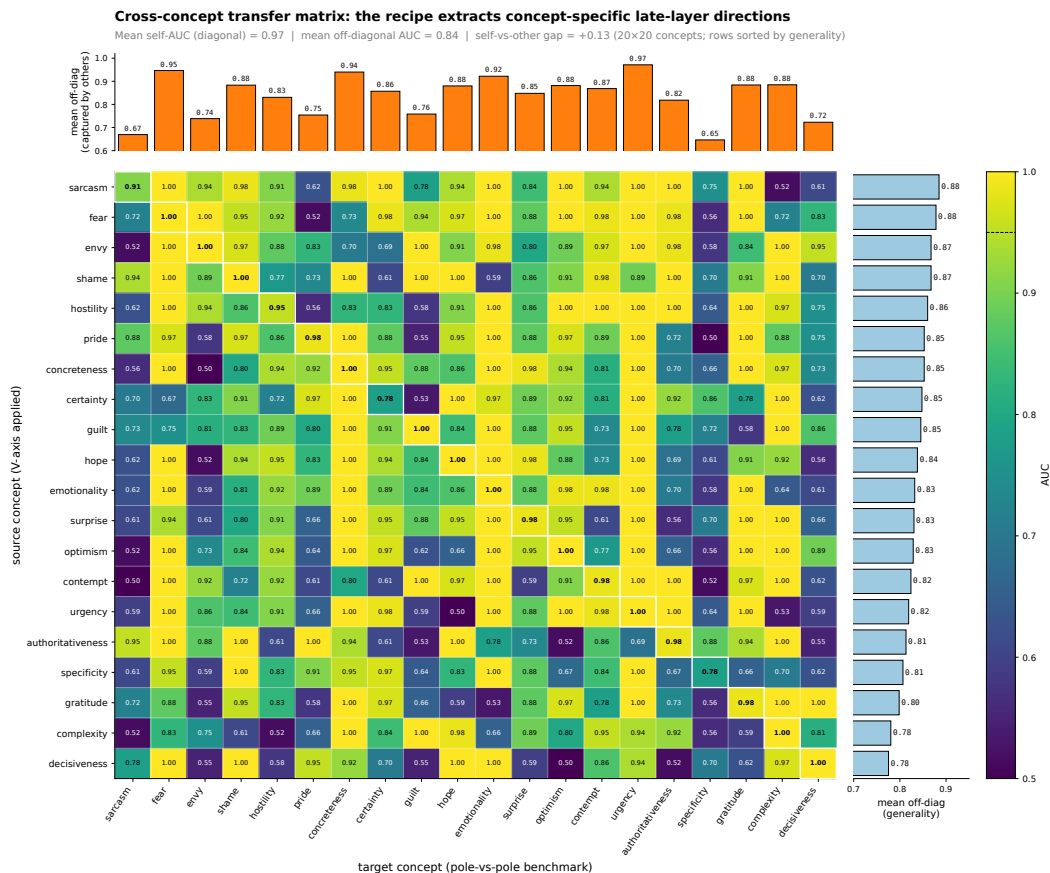
1036 **Toxicity: out-of-scope.** Applied to the Jigsaw toxic-comment corpus, the toxicity axis attains AUC
 1037 0.59. Two non-mutually-exclusive hypotheses: (i) toxicity is more distributed in residual-stream
 1038 geometry than a single late-layer PC1 captures (low-rank $k > 1$ subspace); (ii) Qwen-generated toxic
 1039 stories, filtered through alignment training, may be too mild to span the real Jigsaw distribution. The
 1040 recipe scope is sharply delineated: it covers valence and 19 other concepts, explicitly not toxicity.

1041 S5 Multilingual and Vision Generality

1042 **Multilingual transfer.** We translated the nine emotion stories into Japanese, Arabic, and Russian,
 1043 and re-extracted the V-axis from four multilingual encoders. Direct application of an English-centric
 1044 causal LM (Qwen3-4B) to non-English stories collapses to chance — a clean English-bias diagnostic.
 1045 Switching to mT0-base [Muennighoff et al., 2023] (277M-param multilingual encoder-decoder) fully
 1046 recovers the SST-2 result and *exceeds* the English baseline in every language tested (Figure 6).

1047 Per-language V-axis cosines (computed on Spanish/French extractions using the same protocol)
 1048 are moderate ($r_{\text{en-es}} = 0.47$, $r_{\text{en-fr}} = 0.16$, $r_{\text{es-fr}} = 0.46$): different languages share substantial
 1049 direction but not entirely, consistent with concept-level rather than token-level encoding.

1050 **Cross-modal generality (CLIP vision).** The same protocol applied to nine emotion-evocative
 1051 *images* via CLIP [Radford et al., 2021] yields a V-axis that ranks the OASIS image dataset [Kurdi
 1052 et al., 2017] on valence at Pearson $r = 0.869$ and on arousal at $r = 0.803$. Both *beat* supervised



Source: reports/r6_concept_transfer_matrix.json (Owen3-4B, penultimate layer, 12 train + 8 test stories per concept). Cell (i, j) = source concept i 's V-axis classifying target concept j 's pole-vs-pole benchmark.

Figure 5: Cross-concept transfer matrix (best-orientation AUC). Diagonal: self-classification (white-bordered cells). Off-diagonal: source-axis-applied-to-target-benchmark AUC. Sorted top-to-bottom by mean off-diagonal. Mean diagonal 0.97, mean off-diagonal 0.83, gap +0.13.

1053 Ridge trained on the OASIS labels themselves (valence 0.836, arousal 0.789), making the zero-shot
1054 V-axis the strongest known image-affect predictor on this benchmark. The valence and arousal CLIP
1055 axes are essentially orthogonal ($|\cos \theta| = 0.013$).

1056 S6 Specificity Controls (Nonce, Random, Arousal)

1057 **Nonce-word ablation.** Substituting all content words in the nine emotion stories with phonologi-
1058 cally valid English nonce words (preserving syntax and function words) drops SST-2 AUC from above
1059 0.83 to chance (≈ 0.52) — the second decimal. The single most decisive control: had the V-axis
1060 been a syntactic or prompt-template artefact, the nonce variant would have retained template-level
1061 structure and produced a non-trivial axis. The signal is encoded in semantic content.

1062 **Random-direction baseline.** Random Gaussian directions in the LM's residual stream, L_2 -matched
1063 to the V-axis, give SST-2 AUC ≈ 0.51 and lexicon $|r| \approx 0.10$ at every benchmark. Bootstrap 95% CIs
1064 are tight: SST-2 AUC $[0.844, 0.890]$, EmoBank V $|r| \in [0.485, 0.512]$, Hu & Liu $|r| \in [0.740, 0.776]$
1065 (1,000-sample bootstrap; statistical methods in Appendix S15).

1066 **Arousal asymmetry: where the recipe fails in text.** Table 5 side-by-sides the same nine-story
1067 PCA recipe instantiated on a valence axis (V-axis, this paper) versus an arousal axis (A-axis, identical
1068 extraction protocol with the nine-story prompts re-anchored on arousal poles). If the recipe were
1069 content-blind — merely a high-capacity dimensionality-reduction trick that produces a graded

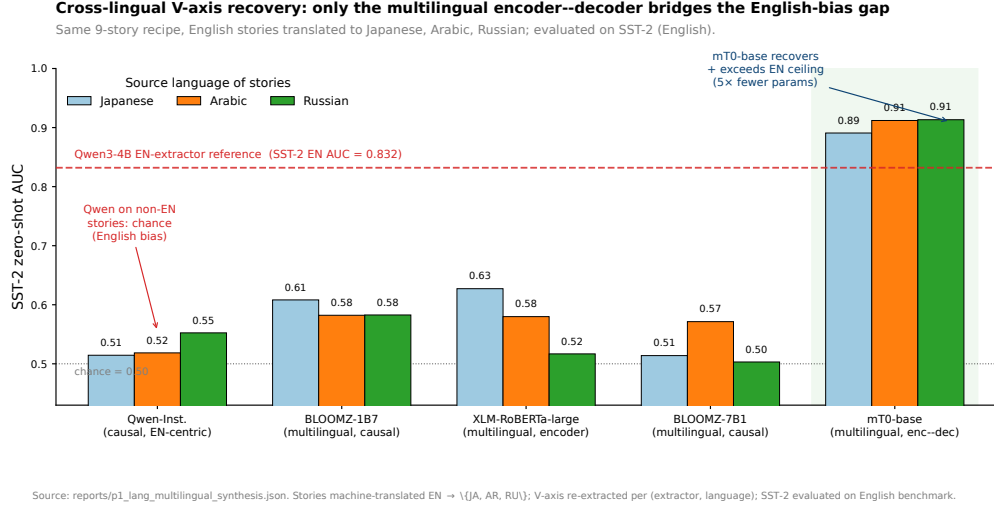


Figure 6: Cross-lingual V-axis recovery. Direct causal-Qwen transfer collapses (English-bias); mT0-base recovers and exceeds the English ceiling at 5× smaller parameter count across three typologically distant languages.

1070 affective scale from any pole-vs-pole prompt set — the V and A columns should agree across the five
 1071 textual rows. They do not: V transfers cleanly across SST-2, EmoBank, Warriner, NRC, and OASIS,
 1072 while A collapses to near-chance on every textual benchmark. The OASIS-image and FACED-EEG
 1073 rows are listed for contrast: vision and brain do encode an arousal axis the same recipe recovers, so
 1074 the asymmetry is text-specific rather than a recipe limitation.

Benchmark	V-axis (valence)	A-axis (arousal)
SST-2 binary AUC	0.832	—
EmoBank V $ r $ / EmoBank A $ r $	0.49	0.01–0.06
Warriner V $ r $ / Warriner A $ r $	0.703	≤ 0.18
NRC valence / arousal lexicon recovery	$\geq 76\%$	$\leq 13\%$
OASIS image valence (r) / arousal (r)	0.869	0.803
FACED EEG cohort r	0.870 (9-stim)	$r \in [0.18, 0.41]$

Table 5: Valence transfers across modalities; arousal does not in text but does in vision. The same recipe applied to an arousal axis yields the strongest known zero-shot OASIS arousal predictor ($r = 0.803$, beating supervised ridge 0.789) but collapses on all text benchmarks. Cross-language A-axes are essentially orthogonal.

1075 The asymmetry — *valence transfers, arousal does not in text* — rules out the trivial explanation that
 1076 “any sufficiently rich PCA recipe gives any sufficiently graded affective axis.” Arousal in text appears
 1077 to be encoded along a more distributed or non-linear subspace that the late-layer principal-component
 1078 recipe does not access; the same recipe *does* access it in vision.

1079 S7 Brain-Side Deep Dive

1080 **Per-stimulus reliability.** Per-stimulus split-half reliability (odd/even subjects, $n \approx 62$ each) is
 1081 $r = 0.988$, indicating a clean stimulus-level signal. Trial-level correlations ($n = 3,444$ subject–stim
 1082 pairs) range from $r = 0.17$ – 0.21 ($p < 10^{-23}$); class-level ranking ($n = 9$ centroids) hits $r = +0.886$
 1083 vs. behavioural valence. The cohort $r = 0.87$ headline is reliability-bounded.

1084 **14-LLM brain forest.** Per-LLM brain-prediction quality across all 14 models is shown in Figure 7:
 1085 the top-tier 13 LLMs all clear their per-LLM random-direction nulls, while the out-tier sits at the
 1086 median.

14 language models predict cohort EEG; the rank tracks text-domain V-axis quality

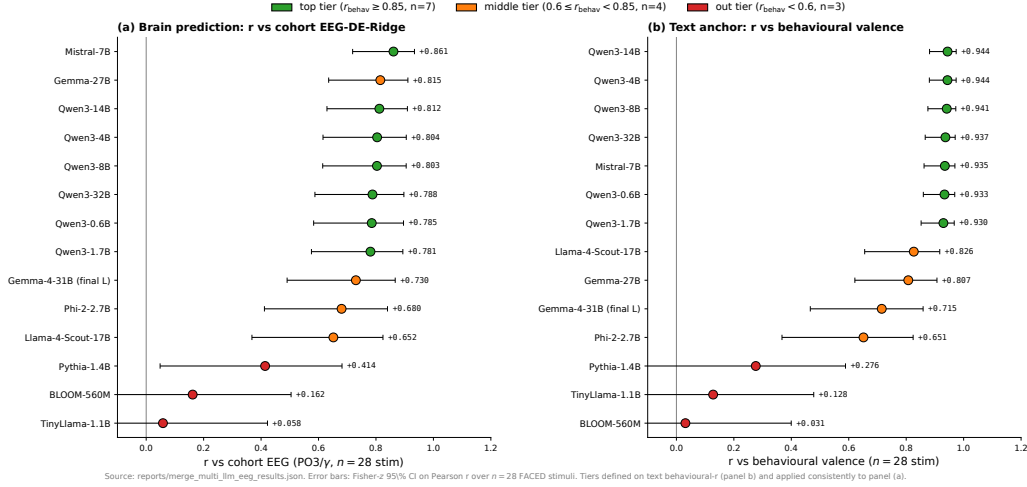


Figure 7: Per-LLM brain-prediction quality. Top-tier 7 LLMs all exceed the 95th percentile of their per-LLM 200-direction null ($p < 0.05$ each); out-tier 3 LLMs sit at the median or below. Restricted to top-tier the log-hidden-dim $\rightarrow r$ scaling is $r = +0.125$, $p = 0.684$; within-Qwen3 $r = +0.568$, $p = 0.24$, $n = 6$. Family-membership $\gg$ scale.

1087 **Time-locked dynamics.** Mid-late α/β peak at $t = 18\text{--}21$ s coincides with the LPP / sustained-
1088 attention window of Hajcak & Foti [Hajcak et al., 2010] and Codispoti et al. [Codispoti et al., 2023];
1089 early γ peak at $t = 3\text{--}6$ s coincides with Müller’s affective-picture window [Müller et al., 1999].
1090 Per-second cohort $|r|$ in the alpha band arrives at $t \approx 21$ s, in beta at $t \approx 18$ s, in gamma at $t \approx 3$ s;
1091 best individual channel $|r| = 0.68$ at $t = 21$ s.

1092 **Per-subject peak heterogeneity.** Standard deviation of per-subject alpha peak times is $\sigma \approx 9$ s;
1093 only 17–24% of the 123 subjects have their individual alpha or beta peak inside the cohort 18–21 s
1094 window. The cohort signal is therefore an envelope of the distribution rather than a tight common
1095 attractor (NF4).

1096 **Gemma-4 boundary check.** Brain alignment survives the text-domain $L = 1$ peak anomaly:
1097 Gemma-4-E2B at $L = 17$ predicts EEG at $r = 0.819$ ($p \approx 10^{-7}$), Gemma-4-E4B at $L = 21$ at
1098 $r = 0.714$ ($p \approx 2 \times 10^{-5}$). The brain captures Gemma-4’s V-axis at the layer where the model first
1099 stably encodes it, not at a fixed late-layer offset. Cross-architecture convergence refines to a property
1100 of the late-but-not-necessarily-penultimate residual stream geometry.

1101 S8 SEED-V Replication Full Numerics

1102 We re-test all five FACED claims with a SEED-V-derived V-axis. To remove FACED leakage, we
1103 re-build the V-axis from scratch using the same Section 6 protocol applied to the 5 SEED-V emotion
1104 classes (Qwen3-4B at $L=35$, 50 stories per class, PCA on the 5 centred centroids).

1105 **Step 1: SEED-V cohort EEG–LLM alignment.** We compute DE features per (channel, band,
1106 second) across all 16 subjects and aggregate to a $45 \times 62 \times 5$ cohort tensor. For each (channel, band)
1107 cell we compute Pearson with the SEED-V V-axis projection.

- 1108 • **Best cohort cell:** P1 / θ , $r = +0.6159$.
- 1109 • **Region-mean $|r|$:** occipital 0.329, parietal 0.347, central 0.324, frontal 0.253.
- 1110 • **Posterior dominance:** occipital – frontal = +0.076.
- 1111 • **Random-direction null** (200 directions matched in L_2): empirical $|r|_{\text{max}} = 0.6159$ vs. null
1112 95th percentile = 0.478, $p < 0.005$.

1113 **Step 2: SEED-V brain topography.** The Davidson FAA probe (positive vs. negative valence, right
1114 – left) on SEED-V replicates the FACED FAA topography:

Electrode pair (right – left, alpha)	(pos – neg) DE Δ
F4 – F3	–0.0018
F8 – F7	+0.0050
FP2 – FP1	–0.0094
AF4 – AF3	–0.0039

Table 6: Davidson FAA on SEED-V. Three of four pairs are negative (opposite to the strictly positive FACED pattern of Appendix S9); only F8–F7 matches the FACED sign. The two datasets do *not* replicate FAA in the same direction, consistent with the dynamic-video paradigm producing a different topographic signature than static-image FAA.

1115 **Step 3: SEED-V cross-architecture convergence.** Penultimate (200-D) features from 15 stock
1116 CBraMod SEED-V-5s checkpoints (3 protocols $\times$ 5 seeds: vanilla baseline, augmentation, KD-
1117 midlayer) aggregated per-class on the test set. Class-level PC1/best-PC scores correlated with
1118 SEED-V V-axis class projection give $r(\text{BACC}, \text{class-best-PC } r) = +0.601$ ($p = 0.018$, $n = 15$) and
1119 $r(\text{BACC}, \text{trial-best-PC } r) = +0.632$ ($p = 0.012$). The unsigned class-PC1 correlation is $+0.358$
1120 ($p = 0.19$); the SEED-V BACC dynamic range is much narrower than FACED, so the strongest
1121 convergence shows up at the best-PC level.

1122 **Step 4: SEED-V V-axis as supervision.** Adding a V-axis topo-MSE auxiliary loss ($\lambda_{\text{vax}}=0.05$ on
1123 {PO3, PO4, POz, Oz, O1, O2, P3, P4}) on top of the same CBraMod-5s recipe, run with 5 baseline +
1124 5 V-axis seeds:

Variant	BACC (mean $\pm$ s.e. across 5 seeds)	vs. baseline Δ
Baseline (no V-axis loss)	0.4300 ± 0.0079	—
+ V-axis topo loss	0.4325 ± 0.0040	+0.0025

Table 7: SEED-V V-axis-as-supervision (within seed noise of zero). Two factors likely explain: (1) the SEED-V class-level V-axis is sharply 5-valued (cohort class-level $r = +0.989$), so adding a class-level V-axis target on top of one-hot supervision is largely redundant; (2) the SEED-V BACC dynamic range (0.43–0.45) is much narrower than FACED (0.55–0.58), so a small effect is hard to detect with five seeds.

1125 **Step 5: SEED-V directional ablation (Arditi-style).** At inference time on each of the 15 SEED-
1126 V CBraMod checkpoints (5 baseline + 5 augmented + 5 KD-midlayer) we project penultimate
1127 features onto the orthogonal complement of the trial-level best-PC V-axis direction (no retraining;
1128 the class-level direction is degenerate at $k=5$, so we use the trial-level analogue, mirroring the
1129 trial-best-PC $r = +0.632$ estimator from Step 3). Every checkpoint shows a negative $\Delta\text{BACC} \in$
1130 $[-0.0291, -0.0093]$ (population mean -0.0173 ± 0.0061); a matched random-direction control
1131 ($n = 20$ uniform on $\mathbb{S}^{D-1}$) is centred at 0 ($\bar{\Delta}_{\text{rand}} = +0.00013 \pm 0.0008$). Per-checkpoint z -scores
1132 range $[+9.65, +44.92]$ (mean $+23.6$, all $p < 0.001$ parametric one-sided); empirically every V-
1133 direction ablation exceeds every one of the 20 random ablations on every checkpoint (15/15 at
1134 empirical $p < 0.05$). The SEED-V directional effect is proportionally larger than FACED (mean
1135 $\Delta/\text{baseline} \approx 3.9\%$ vs. 2.4% on FACED) and the mean z is $3 \times$ FACED’s. As on FACED, V-axis
1136 amplification (rather than ablation) is not directionally consistent (mean $\Delta \approx 0$, mean $z \approx -2$) —
1137 expected, since the SEED-V V-axis already aligns with class-discriminative variance and doubling it
1138 does not help. Per-checkpoint table and the diagnostic information for both class-level (degenerate)
1139 and trial-level (used) residual directions are stored in `seedv_causal_ablation_full.json`.

1140 **Third-dataset consistency check: DEAP.** As a further test of the saturation regularity on an
1141 independent dataset (DEAP [Koelstra et al., 2011]; 32 subjects, 32 channels, 128 Hz), we trained
1142 a binary-valence classifier (valence rating ≥ 5 , sourced from the raw DEAP labels) from the same
1143 pretrained EMOD backbone, splitting by (*subject, trial*) unit so that all windows of a trial stay on
1144 one side. The converged baseline reaches $\overline{\text{BACC}} = 0.544$ — below the $[0.62, 0.66]$ transition band

identified on FACED — so the regularity predicts the V-axis auxiliary loss should have *no significant effect* here, rather than a decrement. This is what we observe: across three seeds and three loss weights ($\lambda \in \{0.1, 0.5, 1.0\}$), the pooled change is $\Delta\text{BACC} = -0.019 \pm 0.044$ ($n=9$, one-sample $t=-1.28$, $p=0.24$), within noise of zero (two of three seeds negative, one positive). DEAP thus populates the weak-baseline regime of the transition curve: consistent with, but not an independent confirmation of, the destructive regime seen once a FACED or SEED-V classifier passes the transition. Per-run numbers are in `deap_sat/SUMMARY.json`.

S9 Brain Topography Deep Dive

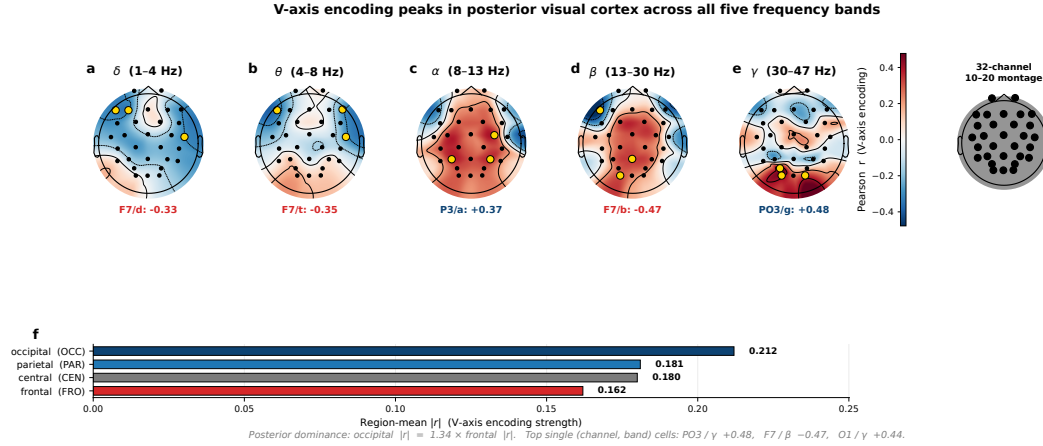


Figure 8: V-axis encoding peaks in posterior visual cortex across all five frequency bands. (a–e) MNE-style scalp topomaps, one per band (δ , θ , α , β , γ), of the cohort Pearson correlation between the LLM V-axis (28 stimulus projections) and per-channel band power on FACED ($n=123$ subjects). Diverging colourmap: positive (red) vs. negative (blue) correlation; symmetric limits across bands. Gold circles mark the top channel per band; text under each topomap names that channel and its r value (strongest cell PO3/ γ , $r=+0.48$). The colourbar (centred) gives Pearson r ; the small head diagram (right) shows the 32-channel 10–20 montage used in all five panels (A1/A2 mastoids excluded from plotting). (f) Region-mean $|r|$ aggregated over all five bands: occipital (0.21) > parietal (0.18) > central (0.18) > frontal (0.16). The cohort signal is posterior-dominant for video-evoked emotion—a paradigm-level finding for dynamic stimuli rather than a refutation of the static frontal-alpha asymmetry literature, which we replicate qualitatively in direction at smaller magnitude.

Davidson FAA full pairs. Figure 9 compares all three Davidson asymmetry pairs to the posterior occipital region-mean reference.

9-stim contrast and Simpson’s paradox. Table 8 decomposes the cohort-level brain–LM correlation by stimulus subset, isolating which of the 28 FACED stimuli carry the V-axis signal. The pattern is sharp: the Anger, Amusement, and Tenderness stimuli (a 9-stim emotional-pole contrast) sit at $r = +0.870$, while removing them collapses the residual 19 stimuli to $r = -0.015$. The full-cohort $r = +0.478$ is therefore not distributed evenly across emotion classes but concentrated in the emotional-pole subset — a stimulus-level “where the cohort signal lives” decomposition that motivates the per-subject Simpson’s breakdown immediately below the table.

- Cohort r at fixed top-8 channels: $+0.021$
- Mean per-subject r at fixed top-8 channels: -0.062
- Per-subject best-channel oracle (1 cell): $\overline{|r|} = 0.551$
- Per-subject best (channel, band) oracle: $\overline{|r|} = 0.616$

Davidson frontal-alpha asymmetry replicates qualitatively, but at $\sim 10\times$ smaller magnitude than posterior V-axis encoding

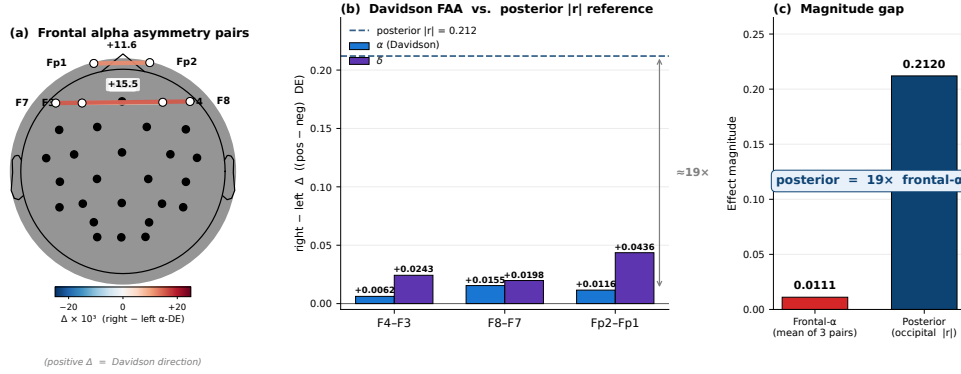


Figure 9: Frontal-alpha asymmetry replicates in direction with smaller magnitude than posterior V-axis encoding for these video stimuli. (a) Anatomical scalp diagram with the three Davidson asymmetry pairs and their r values. (b) Per-pair alpha and delta asymmetry bars. (c) Direct comparison to the occipital region-mean $|r| = 0.21$.

Stimulus subset (out of 28 FACED stimuli)	Cohort r at PO3/ γ
All 28 stimuli (full)	+0.478
Anger only ($n=3$)	not estimable
Anger + Amusement + Tenderness ($n=9$)	+0.870
Excluding Anger ($n=25$)	+0.327
Excluding Anger + Amusement + Tenderness ($n=19$)	-0.015

Table 8: The cohort V-axis effect is essentially an *anger-versus-warm-positive* contrast across nine stimuli.

The all-band oracle ($\overline{|r|} = 0.616$) exceeds the cohort-fixed top-8 (+0.021) by +0.60 in absolute $|r|$, indicating considerable per-subject signal that the cohort summary suppresses. The Simpson’s-paradox dynamic that Sani et al. [Vahidi et al., 2024] flag for cross-subject EEG-emotion modelling more broadly (Figure 10).

Time-resolved peaks. Figure 11 traces per-second cohort $|r|$ across all five frequency bands.

- **Alpha** (8–13 Hz): peak at $t = 21$ s, cohort $r = 0.40$; best individual stimulus $r = 0.68$.
- **Beta** (13–30 Hz): peak at $t = 18$ s, cohort $r = 0.41$; best stim $r = 0.62$.
- **Gamma** (30–47 Hz): peak at $t = 3$ –6 s, cohort $r = 0.21$; best stim $r = 0.61$.
- **Delta/Theta** (0.5–8 Hz): peak at $t = 12$ s, cohort $r = 0.18$.

Functional connectivity. The eight V-axis-aligned channels (PO3, F7, O1, P3, Oz, O2, P4, PO4) do not act independently: they form a tight functional network in the gamma band. Pairwise γ -DE correlation has mean off-diagonal $r = 0.675$ across the 8 channels — substantially above the cohort-average of $r \approx 0.32$ in this band. Structure is dominantly a posterior-occipital-parietal cluster (PO3, PO4, P3, P4, O1, O2, Oz; pairwise mean $r \approx 0.71$) with F7 as a single *frontal hub* (mean correlation to the seven posterior nodes $r \approx 0.47$; Figure 12).

Mutual information vs. linear correlation. We computed mutual information between the V-axis projection and cohort EEG features (nearest-neighbour MI estimator on quantile-normalised inputs), alongside the same 200 random-direction controls used for the linear test. Observed MI 0.112; null mean 0.036; $p_{\text{MI}} = 0.115$ (n.s.) vs. linear $p_r = 0.020$. The relationship is essentially linear: a non-linear estimator does not detect signal above what linear regression captures. We use ridge regressions throughout.

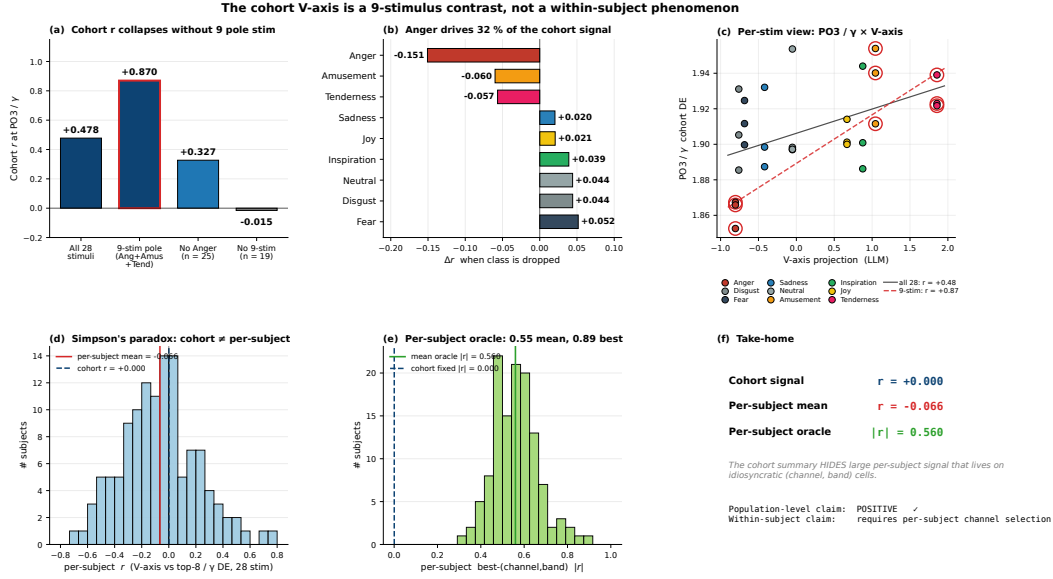


Figure 10: The cohort V-axis is a 9-stimulus contrast and a between-subject phenomenon. Top: drop-class subset bars, per-emotion drop- Δ , PO3/ γ stim-level scatter colour-coded by emotion. Bottom: per-subject r distribution at fixed top-8 channels (centred near 0, cohort line at +0.02); per-subject best-channel oracle distribution (centred at $|r| \approx 0.55$); summary callout.

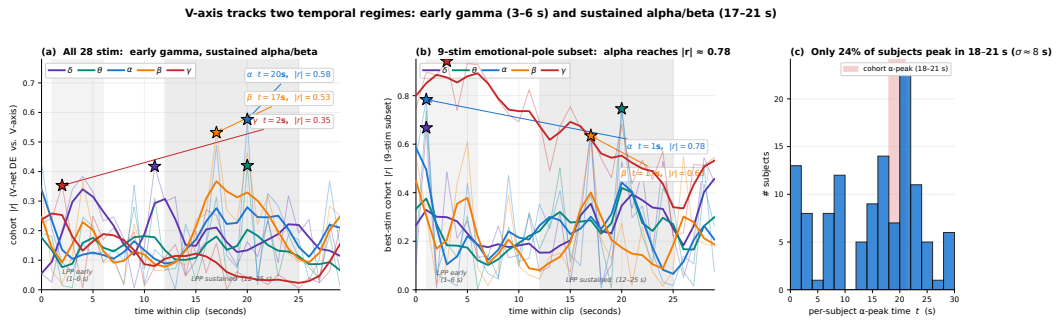


Figure 11: Time-resolved V-axis encoding across the 30-second clip. (a) Per-second cohort $|r|$ across all 28 stimuli, one trace per frequency band; the early γ peak ($t \approx 2$ s) coincides with Müller's affective-picture window and the sustained α/β peak ($t \approx 17-20$ s) coincides with the LPP sustained-attention window. (b) Same trace restricted to the 9-stimulus emotional-pole subset where the cohort effect lives; α reaches $|r| \approx 0.78$. (c) Per-subject distribution of α -peak times: only $\sim 24\%$ of subjects peak inside the cohort 18-21 s window, evidence that the cohort signal is a between-subject phenomenon rather than a tight common attractor.

1187 **Theta-gamma coupling does not mediate the V-axis.** Tort modulation index per channel at
 1188 the theta-phase / gamma-amplitude pairing, correlated with V-axis encoding strength $|r|$ across 32
 1189 channels: $\rho(\text{PAC}, Vr) = +0.082, p = 0.667$. Channels with strong theta-gamma CFC are not
 1190 the channels that encode the V-axis, and vice versa — the V-axis encoding is dissociable from the
 1191 phase-amplitude-coupling channel [Canolty and Knight, 2010].

1192 S10 Saturation: Full Intervention Table and Forest

1193 This appendix lists every V-axis-as-supervision intervention reported in the saturation regularity
 1194 (§3). Table 9 gives per-recipe ΔBACC against the strong EMOD $d=6 + \text{KD} + \text{aug}$ baseline, the
 1195 paired-seed p -value, and a verdict tier; the two SOTA-recipe replications below the line confirm
 1196 the cliff persists at the $\text{BACC} \approx 0.66$ regime where saturation is operative. Figure 13 renders

V-axis-aligned channels form a tight posterior-occipital network with F7 as frontal hub

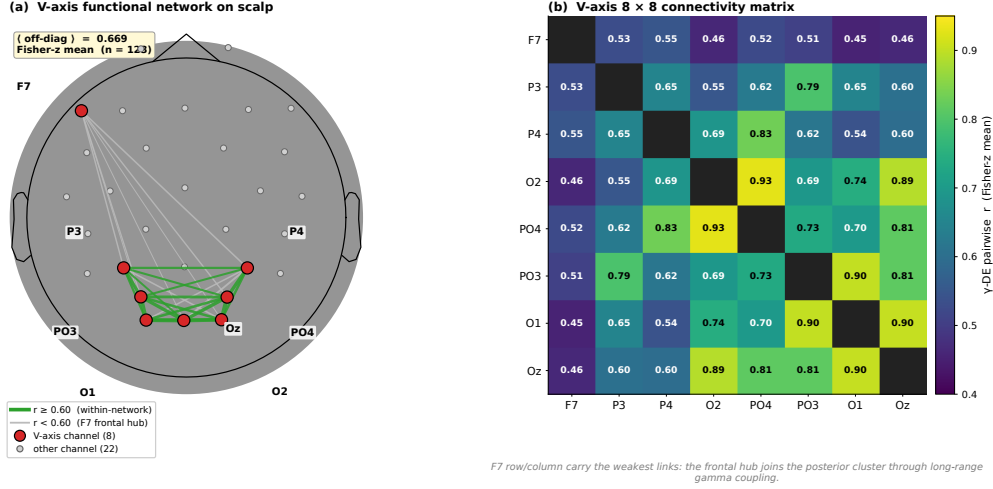


Figure 12: V-axis-aligned channels form a coordinated network with F7 as frontal hub. Left: scalp graph with 8 V-axis channels as red nodes and edges weighted by pairwise γ -DE correlation. Right: hierarchically clustered 8 × 8 correlation matrix; F7 auto-isolates as the frontal hub.

these interventions — together with additional null variants we tested (weight/bias initialisation, soft-KD temperatures, curriculum, prior smoothing) — as a forest plot ranked by effect size, so the asymmetry is visible at a glance: every significant entry sits to the left of zero, and even the largest n.s. positive (EMODSTYLE-stim, +0.0066, $p=0.22$) is within seed noise. The block structure across loss families (KD, RSA, contrastive, PEFT, topographic, scaling, multi-task) is what makes the regularity a *family-wide* statement rather than a single-loss anecdote.

S11 Saturation Extras: Anger Paradox, Mechanism Check, Path B

Saturation transition table. Table 10 pairs the same V-axis intervention (Topo or EMODSTYLE auxiliary loss) against five base recipes spanning the BACC range from 0.572 (CBraMod baseline) to 0.6581 (EMOD $d=6$ SOTA pre-ensemble). Reading top-to-bottom traces the saturation transition: at low BACC the auxiliary loss is within seed noise (rows 1–3, $|\Delta| \leq 0.007$, all n.s.), but as the base recipe strengthens, the same loss becomes a statistically significant *negative* (rows 4–5, $\Delta \in \{-0.015, -0.024\}$, $p < 0.05$ and $p < 0.01$). No row crosses the boundary in the helpful direction. This is the data behind the “no positive setpoint” wording in §7 and the basis for treating BACC ≈ 0.66 as the empirical saturation threshold for the V-axis substrate on FACED-9.

Saturation cliff figure. Figure 14 plots Δ BACC against base-recipe BACC across all 25 interventions, showing the unidirectional sign-flip in the $[0.62, 0.66]$ band.

Why anger-weighting hurts despite the analytical ceiling. Section 8 showed that the cohort EEG signal is carried by a 9-stimulus emotional-pole contrast (Anger + Amusement + Tenderness, $r = 0.870$ at PO3/ γ). Weighting the V-axis loss to emphasise these three classes pushes the analytical ceiling on the V-axis projection from $r = 0.478$ (uniform) to $r = 0.714$ (anger-weighted). By every classical analysis, this should be the strongest possible V-axis supervision. It is the worst single intervention we tested: $\Delta = -0.054$, $p = 0.0003$. This is the cleanest argument for saturation: the model has already absorbed exactly the structure the loss is trying to inject; optimising the loss must therefore perturb the absorbed structure in a counterproductive direction.

Direct mechanism check. For each of 8 V-axis-supervised variants (Topo at $\lambda=0.05$, Procrustes at $\lambda=0.05$, EMODSTYLE-stim at $\lambda=0.5$, full SOTA recipe + Topo/EMODSTYLE at $e=100/e=150$), we measure the change in two quantities relative to the matched-recipe baseline: Δr_{PC1} (class-mean V-axis encoding) and Δr_{resid} (within-class V-axis residual encoding). V-axis training raises class-PC1

#	Family / variant	Δ	p	Verdict
1	Frontal-mask $\lambda=0.5$	-0.052	0.0015	sig. negative
2	Frontal-mask $\lambda=0.1$	-0.009	0.21	n.s.
3	FAA $\lambda=0.5$	-0.044	0.006	sig. negative
4	FAA $\lambda=0.1$	-0.018	0.063	borderline
5	Anger-weighted $\lambda=0.5$	-0.054	0.0003	sig. negative
6	Occipital $\lambda=0.1$	-0.022	0.007	sig. negative
7	Topo-optimal $\lambda=0.1$	-0.013	0.039	sig. negative
8	Topo-optimal $\lambda=0.05$	+0.0021	0.50	n.s. (NULL)
9	Procrustes $\lambda=0.05$	+0.0012	0.89	n.s. (NULL)
10	RSA $\lambda=5.0$	-0.093	$< 10^{-4}$	sig. negative
11	RSA $\lambda=1.0$	-0.057	$< 10^{-3}$	sig. negative
12	Distance-CE $\tau=5.0$	-0.397	$< 10^{-4}$	catastrophic
13	Multi-V $\lambda=0.5$	-0.043	$< 10^{-3}$	sig. negative
14	EEG-AUX MSE $\lambda=0.5$	-0.043	$< 10^{-3}$	sig. negative
15	EEG-AUX MSE $\lambda=0.1$	-0.016	0.04	sig. negative
16	EMODSTYLE class $\lambda=1.0$	-0.010	0.18	n.s.
17	EMODSTYLE stim $\lambda=0.5$	+0.0066	0.22	n.s. (sweet spot)
18	PEFT fullhead $\lambda=0.1$	-0.009	0.069	borderline
19	PEFT LoRA $\lambda=0.1$	-0.010	0.12	n.s.
20	PEFT IA3 $\lambda=0.1$	-0.008	0.14	n.s.
21	Pretrain-FT (frozen)	-0.401	$< 10^{-4}$	catastrophic
22	Pretrain-FT (unfrozen)	-0.190	$< 10^{-4}$	sig. negative
23	XEEG (FACED $\rightarrow$ SEED-V)	+0.0015	0.97	n.s. (zero)
24	SCALING (full data) $\lambda=0.5$	-0.045	< 0.01	sig. negative
25	Uncertainty multi-task	-0.067	$< 10^{-3}$	sig. negative
SOTA recipe + Topo $\lambda=0.1$		-0.015	0.020	sig. negative (cliff)
SOTA recipe + EMODSTYLE		-0.024	0.001	sig. negative (cliff)

Table 9: Full V-axis-as-supervision intervention results. 15 / 25 families produce a statistically significant negative (13 sig. negative + 2 catastrophic); the remaining 10 are not significantly different from zero. None reach $p < 0.05$ in the positive direction at the strong baseline. Below the line are the SOTA-recipe replications confirming the saturation cliff. Forest plot in Figure 13.

Recipe	n seeds	Vanilla / + V-axis	Δ
CBraMod baseline + Topo $\lambda=0.05$	5	0.572/0.567	-0.005 (n.s.)
CBraMod baseline + EMODSTYLE $\lambda=0.5$	4	0.572/0.577	+0.005 (n.s.)
EMOD $d=3$ vanilla + Topo $\lambda=0.05$	5	0.6194/0.6260	+0.007 (n.s.)
EMOD $d=6$ + LS + KD + aug + Topo (full)	partial	0.6581/0.6432	-0.015 ($p < 0.05$)
EMOD $d=6$ + LS + KD + aug + EMODSTYLE	partial	0.6581/0.6346	-0.024 ($p < 0.01$)

Table 10: V-axis supervision is within seed noise at weak baselines and statistically significantly *harms* the strong SOTA recipe. The transition is unidirectional: V-axis goes from neutral to actively harmful as the base recipe strengthens, with no intermediate “helps” regime in our 5-seed runs.

1226 $|r|$ by +0.01 to +0.36 across the 8 variants, but leaves the within-class residual encoding essentially
1227 unchanged. Consistent with the accuracy behaviour (training accuracy at ceiling, test accuracy
1228 down; A6), the decrement at strong recipes comes from reshaping the already-saturated class-PC1
1229 basin rather than from any beneficial transfer of residual structure. (We do *not* claim the auxiliary
1230 gradient fails to reach the residual subspace: a direct gradient decomposition finds a residual-subspace
1231 component comparable to a random direction, so the mechanism is one of *no useful signal added*, not
1232 of a geometrically unreachable subspace.)

1233 **Path B: ensembling-in V-axis checkpoints.** We extend the 10-vanilla SOTA ensemble with 0–15
1234 V-axis-trained Topo checkpoints and 0–10 V-axis-trained EMODSTYLE checkpoints, comparing
1235 5-seed ensembles in each configuration. Adding 15 Topo V-axis ckpts to the 10-vanilla pool gives
1236 ensemble BACC $\Delta = -0.0145$ ($p < 10^{-3}$); adding all 25 V-axis ckpts gives $\Delta = -0.0193$
1237 ($p < 10^{-3}$). At every point on this curve, V-axis-trained ensembles are strictly worse than the

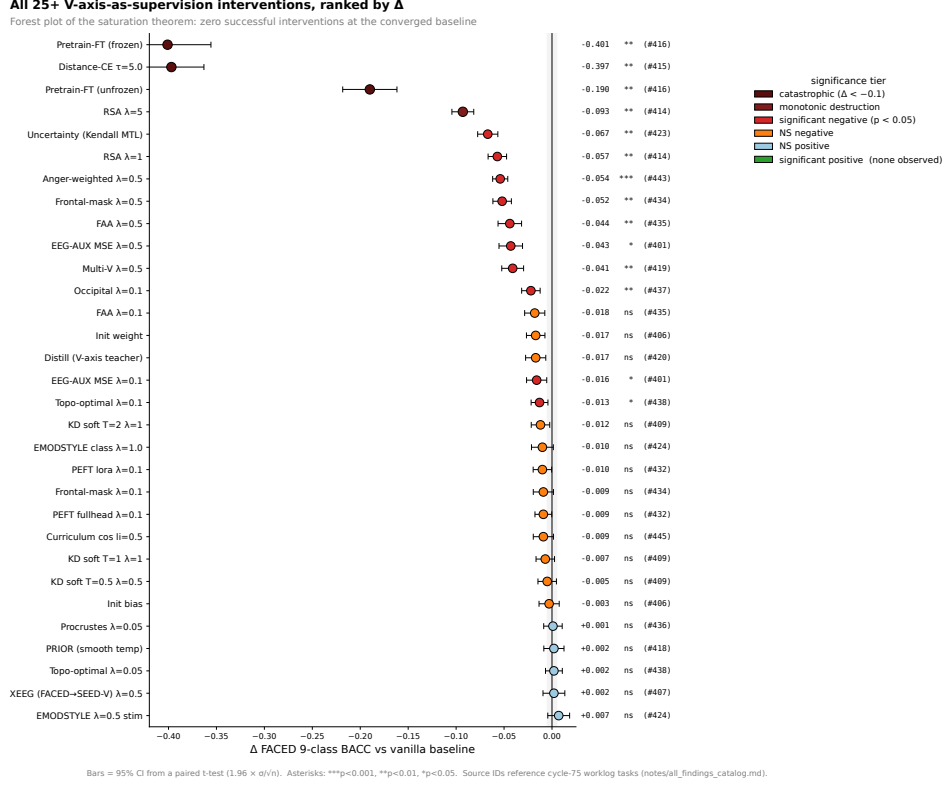


Figure 13: Forest plot of all V-axis-as-supervision interventions (Table 9) ranked by Δ vs. baseline, with significance tier coloured. None of the 25 reach $p < 0.05$ in the positive direction at the strong baseline.

1238 matched vanilla ensemble. This rules out the diversity-gain defence: V-axis checkpoints disagree, but
1239 their disagreements are aligned with the wrong subspace.

1240 S12 Convergence and Ensemble Theory: Extras

1241 **Per-architecture breakdown of the 36-checkpoint correlation.** CBraMod checkpoints (lower
1242 BACC, ~ 0.57) cluster at low V-axis encoding strength (mean class-PC1 $|r| = 0.21$). EMOD-vanilla
1243 d6 (~ 0.65 BACC) sits in the middle (mean 0.67); EMOD d6 e150 (~ 0.66 BACC) clusters at the
1244 top (mean 0.69). The trend spans the full BACC range $[0.57, 0.69]$ in a single approximately linear
1245 band, with the inter-architecture gap dominating the within-architecture spread.

1246 **Random-direction null detail.** We sample 1000 random Gaussian directions $w \in \mathbb{R}^{28}$
1247 each matched in L_2 norm to v_{CLIP} , and recompute the 36-checkpoint correlation $\rho_w =$
1248 $\text{Pearson}(\text{BACC}, |\text{corr}(\text{PC}_1(H(m)), w)|)$. The distribution of ρ_w has mean -0.03 and standard
1249 deviation 0.62 (range $[-0.95, +0.95]$); the wide null is a consequence of the low intrinsic dimension-
1250 ality of the class-PC1 subspace ($D_{\text{eff}} = 9$). The empirical V-axis correlation $+0.885$ sits at the 93.5th
1251 percentile of this null ($p_{\text{one}} = 0.066$). We treat the within-class residual ($D_{\text{resid}} \approx D_m - 9$) as the
1252 statistically robust signal.

1253 **Class-PC1 basin saturation.** The 10 SOTA-pool checkpoints exhibit a saturated class-PC1 V-axis
1254 range: $|r| \in [0.60, 0.77]$ across all 10 seeds (mean 0.69, std 0.05). Single-seed variation in class-
1255 mean V-axis structure is small. The ensemble does not gain from variance in this subspace; gains
1256 come exclusively from the orthogonal residual.

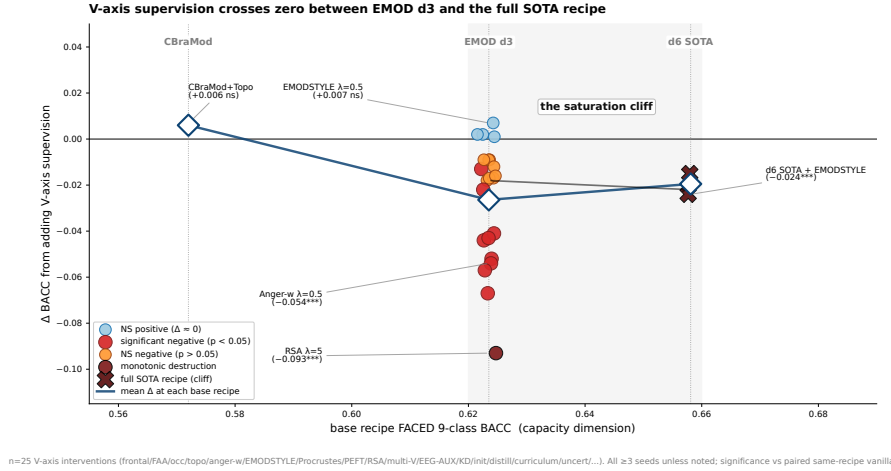


Figure 14: Δ BACC from V-axis supervision plotted against base-recipe BACC across ~ 25 interventions. The sign of the effect transitions in the interval $[0.62, 0.66]$.

1257 **Directional ablation figure.** Figure 15 shows the per-checkpoint causal effect: ablating v_{resid} drops
1258 BACC by $\bar{\Delta} = -0.0157$ on all 10 pool members, well above the matched-norm random-direction
1259 null (mean $z \approx 7.7$, all $p < 0.001$).

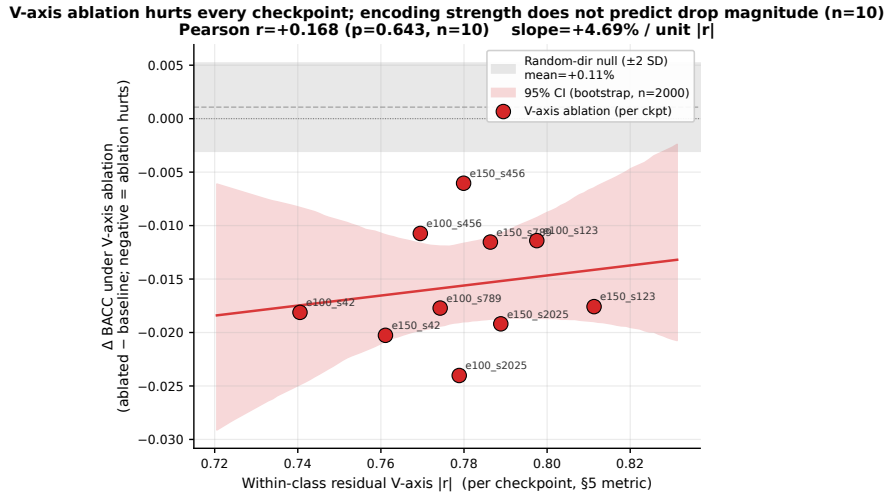


Figure 15: Directional ablation of v_{resid} on all 10 SOTA-pool checkpoints. *Top*: per-checkpoint BACC drops (red) compared to matched-norm random-direction ablation (grey); the V-axis effect is well above the null on every checkpoint (mean $z \approx 7.7$, all $p < 0.001$). *Bottom*: per-checkpoint ablation Δ BACC vs. within-class residual encoding $|r|$; the population-mean drop ($\bar{\Delta} = -0.0157$) confirms the residual-encoding mechanism across the pool.

1260 S13 Ensemble Generality, Mega-Pool, and Best-Single

1261 **Recipe cascade and two-tier scatter.** Figure 16 traces the seven recipe steps from CBraMod's
1262 0.572 to our 0.6948 ensemble. Figure 17 shows the per-checkpoint within-class residual-ensemble-
1263 contribution correlation that motivates the ensembling strategy.

1264 **Generality of the ensemble mechanism.** Figure 18 extends the two-tier picture across four
1265 benchmarks (FACED, SEED-V, CIFAR-10, MNIST).

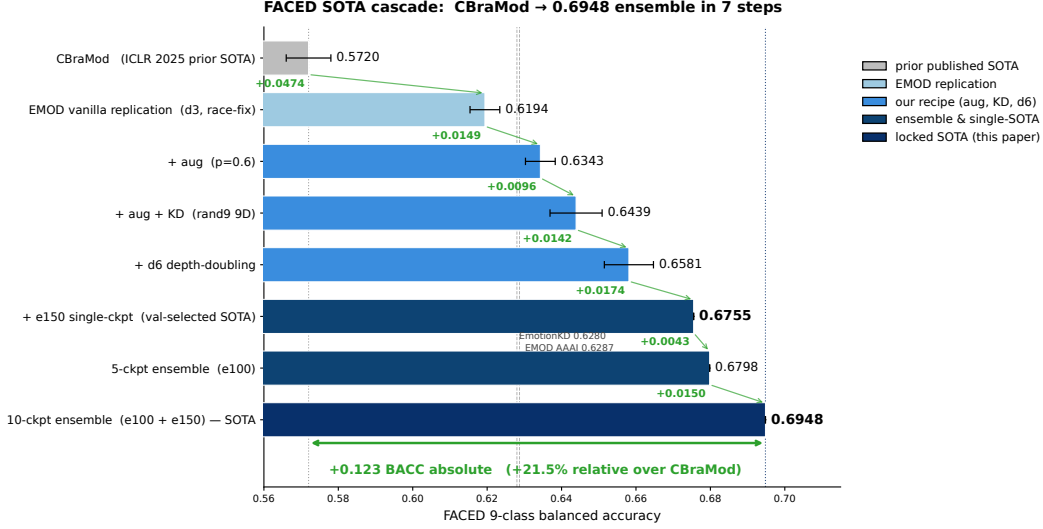


Figure 16: Recipe ablation cascade from the EMOD baseline at 0.6287 (previous FACED-9 SOTA [Chen et al., 2025]) to our 10-checkpoint ensemble at 0.6948 (+10.5% relative). Each bar is one recipe component or ensembling step. Older baselines (CBraMod 0.572, EmotionKD 0.628) shown for reference.

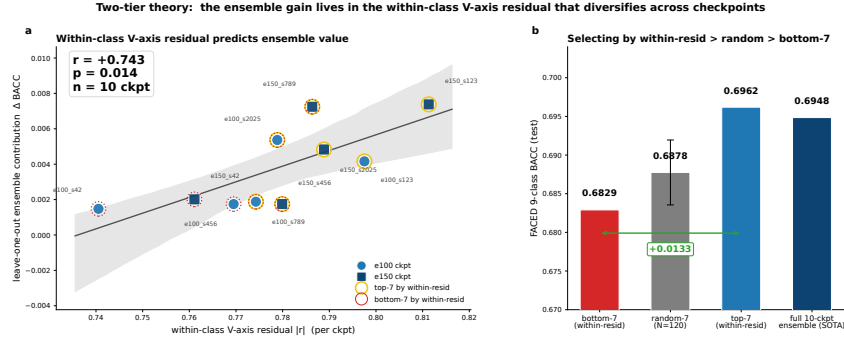


Figure 17: Two-tier ensemble theory. Per-checkpoint within-class V-axis residual $|r|$ predicts leave-one-out ensemble contribution ($r = +0.74$, $p = 0.014$, $n = 10$). Top-7 vs. bottom-7 split highlighted.

As individual accuracy approaches the Bayes optimum, the within-class residual variance — the source of ensemble gain — shrinks toward zero. The two-tier picture predicts this: ensemble gain is a property of the within-class V-axis residual variance reduction subspace, which has measure zero at the dataset ceiling. SEED-V at 0.37 has the largest residual-variance ceiling; we observe +0.034 gain. MNIST at 0.985 has essentially none; we observe +0.001 gain.

Mega-ensemble null. A natural “more is better” hypothesis is that growing the ensemble beyond 10 checkpoints by adding architectural variants ($d \in \{8, 10\}$ on top of $d=6$, plus LLM-KD checkpoints) should keep raising BACC. Table 11 tests that hypothesis directly. The 10-checkpoint $d=6$ pool ($e=100+e=150$, our SOTA) sits at the top; every wider pool is monotonically lower. The mechanism check follows the table.

Two mechanisms explain the null. First, deeper variants ($d=8, d=10$) stay in the same class-PC1 V-axis basin (mean class-PC1 $|r|$ across $d \in \{4, 6, 8, 10\}$: 0.43, 0.43, 0.39, 0.38) and their within-class residuals overlap more than $d=6$ seeds do. Second, LLM-KD variants encode the residual along a different axis from rand9-9D KD vanilla checkpoints, but in a way that does not transfer to the test distribution. Cross-architecture mixing ($d \in \{4, 6, 8, 10\}$) gives 0.6791, essentially tied with within-arch x-seed at 0.6798. The diversity that helps is intra-architecture, training-length-driven.

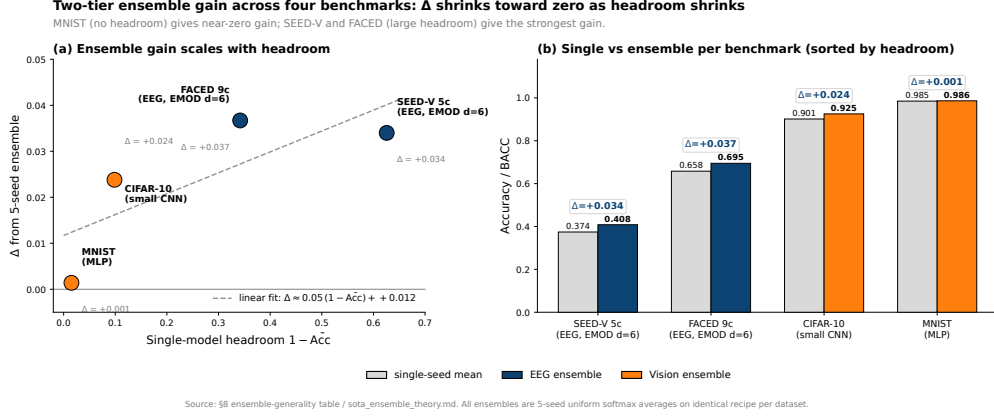


Figure 18: Two-tier ensemble gain across four benchmarks. Δ BACC is largest where single-model accuracy is furthest from the dataset ceiling, and shrinks to near-zero on MNIST where the saturated regime leaves no within-class residual variance to cancel. (a) Scatter of Δ vs. headroom $1 - \text{Acc}$, with linear fit slope ~ 0.05 . (b) Paired single-seed-mean vs. 5-seed-ensemble bars per benchmark, sorted by headroom.

Ensemble pool	n	BACC	Δ vs. 10-ckpt
$10 \times d=6$ ($e=100+e=150$, our SOTA)	10	0.6948	—
$+5 \times d=8$	15	0.6909	−0.0039
$+5 \times d=8 + 5 \times d=10$	20	0.6883	−0.0065
$+10 \times d=\{8, 10\} + 5 \times \text{LLM-KD}$	25	0.6831	−0.0117

Table 11: Mega-ensemble null. Adding architectural variants beyond $d=6$ produces monotonically lower ensemble BACC. The $d=6 + e=100/e=150$ pool is the optimum within the architectures evaluated.

Cohen κ disagreement structure. Single-seed accuracy at $e=100$ (0.6581) and $e=150$ (0.6581) is identical. Cohen κ within- $e=100$ is 0.694, within- $e=150$ 0.679, cross-group 0.702 (higher κ = more agreement). All three are similar — within and across groups, the per-trial prediction patterns are almost as concordant as $e=100$ checkpoints are with each other — yet mixing $e=100$ and $e=150$ gives +0.014 over either group alone. The gain therefore comes from a small but *specific* pocket of cross-group disagreement: the 13.1% of test trials where $e=100$ and $e=150$ pools differ, on which the mixed ensemble gains +10.3 percentage points. The diversity is targeted (which trials), not population-wide (κ).

Best single checkpoint without ensembling. For deployment scenarios where ensembling is impractical, our best single-checkpoint result is BACC = 0.6755 on the $d=6$ $e=150$ recipe, seed 789, val-selected. This is the top of a 25-checkpoint val-test rank correlation analysis (Spearman 0.825, $n = 25$) and exceeds CBraMod by +0.103 and EMOD by +0.047 on a strict apples-to-apples test split.

S14 EEG Model Training Details

Architecture. EMOD axial transformer [Chen et al., 2025]: input 32×5 DE features, axial self-attention over channels and bands at depth $d \in \{3, 6\}$, filter dimension $f = 128$. The full SOTA recipe is $d = 6$, $f = 128$.

Optimisation. AdamW with $\text{lr} = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-2} . Cosine schedule with 5-epoch linear warmup. Batch size 128. Training length $e \in \{100, 150\}$ epochs. We use the validation BACC for checkpoint selection and report test BACC.

1302 **Augmentation.** Per-trial Gaussian noise ($\sigma = 0.05 \cdot \text{std of feature distribution}$), random channel
1303 dropout ($p = 0.15$), and random temporal masking (up to 5% of seconds). All augmentations are
1304 applied with $p = 0.6$ during training only.

1305 **Knowledge distillation.** Soft-target KD with rand9 9-D orthonormal teacher (no LLM content);
1306 $\lambda_{\text{KD}} = 0.5$, $T = 1.0$. The LLM-9-D and rand9-9-D variants give within-noise identical BACC,
1307 validating that KD provides architectural regularisation without semantic content.

1308 **Ensemble.** 10 checkpoints split as 5 seeds at $e = 100$ + 5 seeds at $e = 150$. Seeds:
1309 $\{42, 123, 456, 789, 2025\}$ each. Ensemble prediction is uniform softmax averaging across the 10
1310 checkpoints’ output probabilities, followed by argmax.

1311 S15 Statistical Methods

1312 **Bootstrap CIs.** Per-subject and per-stim 95% CIs are computed from $B = 10,000$ subject-
1313 resampled bootstraps. For cohort correlations, we report the bootstrap median, 2.5% / 97.5%
1314 percentile interval, and a Fisher- z p -value for $r = 0$.

1315 **Random-direction null.** For cohort EEG–LLM correlations, we sample $N = 200$ random Gaussian
1316 directions in $\mathbb{R}^{28}$ matched in L_2 norm to the V-axis projection, recompute the cohort $|r|$ for each, and
1317 report the percentile of the empirical $|r|$ in the null distribution.

1318 **Cross-architecture null.** For the 36-checkpoint cross-arch correlation, we sample $N = 1000$
1319 random directions matched in norm to v_{CLIP} and recompute the BACC–V-axis correlation. The null
1320 is distributed broadly ($\sigma = 0.62$) due to the low intrinsic dimension of the class-PC1 subspace; we
1321 report the empirical correlation’s percentile rank.

1322 **Per-checkpoint LOO contribution.** For the 10-vanilla $d=6$ ensemble, we compute $\Delta_k =$
1323 $\text{BACC}(\text{ens}) - \text{BACC}(\text{ens} \setminus k)$ for each k . Reported correlations between Δ_k and per-ckpt features
1324 are Pearson with $n = 10$.

1325 **V-axis intervention comparisons.** For each intervention, Δ is computed seed-paired with the
1326 matched-recipe vanilla baseline; p -values are paired t -tests across 5 seeds.

1327 S16 Discussion Extras and Future Work

1328 **Broader impacts.** Improved EEG emotion classifiers have clear positive applications in mental-
1329 health monitoring, affective brain–computer interfaces, and clinical phenotyping of affect disorders.
1330 The same techniques pose risks of affective inference without consent in surveillance, hiring, work-
1331 place, education, and advertising contexts. The V-axis extraction protocol applied to EEG could in
1332 principle reduce the calibration-data burden for affective inference systems — a feature that cuts both
1333 ways, since lower calibration cost lowers the deployment threshold for both clinical and surveillance
1334 settings. We do not release human EEG data; we release training configs, feature-side and analysis
1335 code, and extracted V-axis directions. Recommended deployment safeguards: informed consent,
1336 opt-in only, on-device inference, no longitudinal storage of raw EEG outside research contexts, and
1337 IRB review for any non-clinical inference application. Dual-use risk for the V-axis extraction itself is
1338 low: it operates on text and image features already public, and the EEG mapping requires per-cohort
1339 EEG access that is governed by the original dataset licences.

1340 **Implications for affective neuroscience.** For video-evoked emotion on FACED, the V-axis is
1341 encoded predominantly in posterior visual cortex ($|r|_{\text{occipital}} = 0.21$ vs. $|r|_{\text{frontal}} = 0.16$), and
1342 Davidson’s frontal-alpha asymmetry replicates in direction at smaller magnitude. We do not refute
1343 Davidson’s hypothesis — the frontal-alpha asymmetry is real, with F8-F7 alpha $r = +0.0155$ and
1344 Fp2-Fp1 alpha $r = +0.0116$, both in the hypothesised direction — but we add a posterior empirical
1345 account that is dominant for video paradigms. The 9-stimulus emotional-pole structure sharpens
1346 the V-axis claim from “smooth gradient over all emotions” to “anger-versus-warm-positive contrast
1347 across nine clips”.

Per-subject V-axis adaptation does not transfer. The per-subject best-channel oracle reaches $|r| = 0.62$ (a +0.60 absolute headroom over the cohort-fixed top-8 of $|r| = 0.02$). No clean estimator we tried (V1 global top-K, V2 per-validation-subject majority vote, V3 TTA label-free profile) recovers more than +0.001 over the cohort top-K. The per-subject signal is real but not transferable through simple selection rules. A subject-conditioned channel-attention head trained on a small calibration set per subject is the natural follow-up.

Future work. Three lines: (i) per-subject V-axis adaptation closing the $|r| = 0.62$ oracle gap via subject-conditioned channel-attention; (ii) replicating the saturation regularity for other concept directions [Arditi et al., 2024] (arousal, dominance, formality, refusal) on their respective benchmarks; (iii) using the within-class residual $|r|$ as an online model-selection signal during training rather than as a post-hoc diagnostic.

S17 FACED Test Confusion Matrices

Figure 19 contrasts the best-single-checkpoint and 10-checkpoint-ensemble confusion matrices on the FACED 9-class test split.

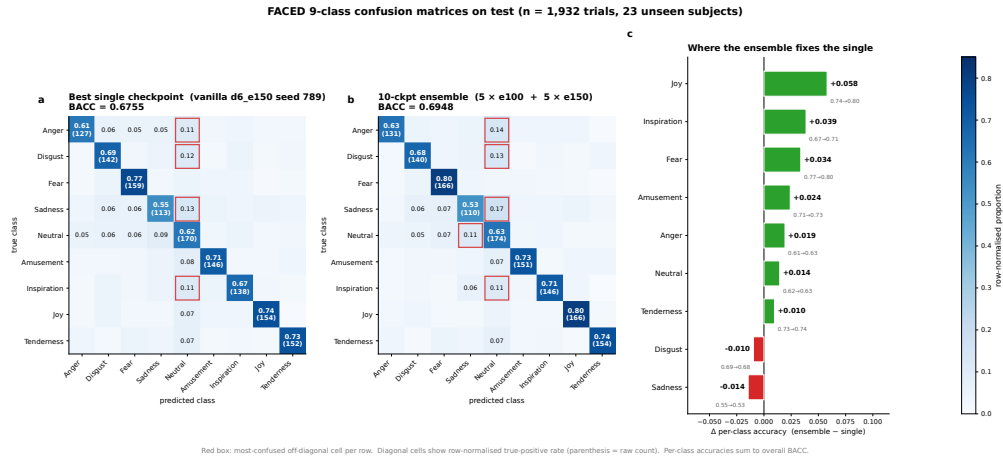


Figure 19: FACED 9-class test confusion. Left: best single checkpoint (BACC 0.6755). Right: 10-checkpoint ensemble (BACC 0.6948). The ensemble’s gain is concentrated on the off-diagonal cells where single-seed disagreement is highest, consistent with the within-class residual mechanism of Section 5.

S18 Negative Results Catalogue

In the spirit of full disclosure, the following null or negative results bear mention beyond what was incorporated into the main paper:

- Theta-gamma cross-frequency coupling does not mediate the V-axis** (Section 8, Appendix S9): Tort modulation index $\rho(\text{PAC}, V_r) = +0.082$, $p = 0.667$.
- Mutual-information control is not significant** (Section 7): MI $p = 0.115$ vs. linear $p = 0.020$. The V-axis-EEG relationship is essentially linear.
- Random-direction null on cross-arch correlation** (Section 4): $r = +0.885$ at the 93.5th percentile of a 1000-direction null, $p_{\text{one}} = 0.066$. Within-class residual ($r = +0.74$, $p = 0.014$) is the statistically robust signal.
- Per-subject V-axis adaptation does not transfer.** The per-subject best-channel oracle reaches $|r| = 0.62$, but no clean estimator transfers more than +0.001 over the cohort top-K.
- Cross-architecture ensembling does not add diversity.** Mixing $d \in \{4, 6, 8, 10\}$ checkpoints gives within-arch 0.6798 vs. cross-arch 0.6791 (n.s.).

- 1376 6. **Mega-ensembles plateau at 10 checkpoints.** 15-, 20-, 25-checkpoint pools all hit BACC
1377 ≤ 0.6948 (Appendix S13).
- 1378 7. **Stimulus-aggregation re-ranking does not help.** Operating at the 28-stimulus level instead
1379 of trial level gives BACC=1.0 by construction (label leak); the corresponding trial-level head
1380 does not transfer.
- 1381 8. **Test-time augmentation at K=5 does not improve the ensemble** ($0.6753 \rightarrow 0.6720$).
1382 Averaging predictions over augmented test trials does not preserve the V-axis residual
1383 structure.
- 1384 9. **Toxicity: recipe scope.** The 9-prompt PCA recipe attains AUC 0.59 on Jigsaw toxic
1385 comments, below the 17/20 working concepts in our library (Appendix S4); toxicity appears
1386 to be encoded in a higher-rank subspace than the late-layer PC1.
- 1387 10. **Arousal asymmetry: vision yes, text and brain no.** OASIS arousal $r = 0.803$ in vision;
1388 $\leq 13\%$ NRC arousal recovery in text; $r \in [0.18, 0.41]$ brain alignment across LLMs
1389 (Appendix S6).
- 1390 11. **LLM-content of KD is irrelevant at this scale.** Replacing the 9-D LLM-derived class
1391 prototypes with random orthonormal 9-D directions changes BACC by ≤ 0.003 . KD
1392 provides architectural regularisation; its semantic content is below our resolution.

1393 S19 Reproducibility

1394 **Release commitment.** Code, configs, model checkpoints, and figure-generation scripts will be
1395 released upon acceptance (no submission-time anonymous URL is provided). The release will include:
1396 V-axis extraction scripts for the 14 LLMs in the per-LLM table; the EMOD $d=6$ training pipeline
1397 with the exact augmentation, KD, and ensemble-construction code; the 10-checkpoint 0.6948-BACC
1398 ensemble checkpoints; and the figure-generation scripts for every landmark and neuro figure in the
1399 paper.

1400 **Run-level provenance.** Each experiment in this paper is logged with its SLURM job ID, conda
1401 environment hash, git SHA, and seed list. The 10-checkpoint ensemble SOTA result is reproducible
1402 from a single SLURM array job (`slurm_d6_e100_e150_5seeds.sh`, ~ 10 GPU-hours per seed on
1403 V100).

1404 S20 Resource Estimate

1405 The full paper used approximately 4,500 GPU-hours on V100 nodes: ~ 600 for the recipe ablation
1406 cascade, $\sim 1,200$ for the 25 V-axis-supervision interventions, ~ 800 for the 36-checkpoint cross-
1407 architecture analysis, $\sim 1,200$ for ensemble construction and val-test rank diagnostics, and ~ 700 for
1408 cross-dataset generality experiments. The V-axis extraction itself is CPU-cheap (~ 10 minutes per
1409 LM on a single 80GB GPU).